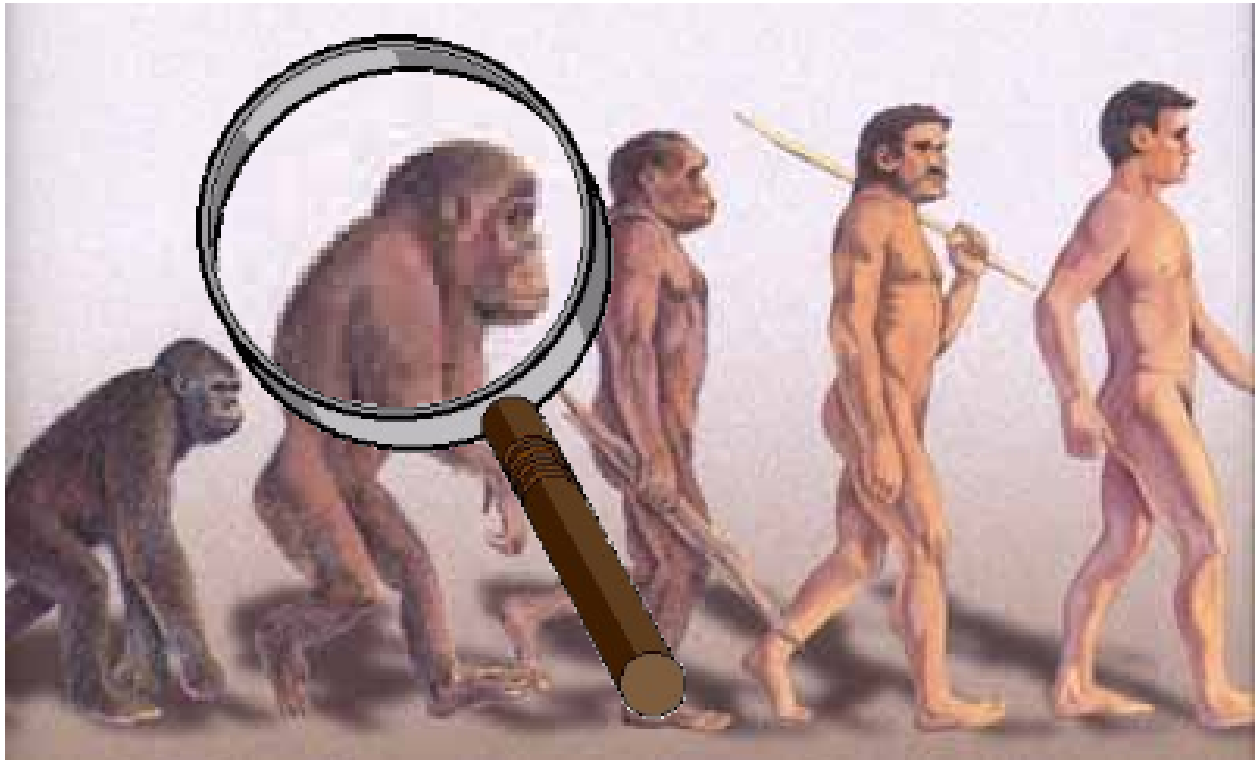
Computational Biology

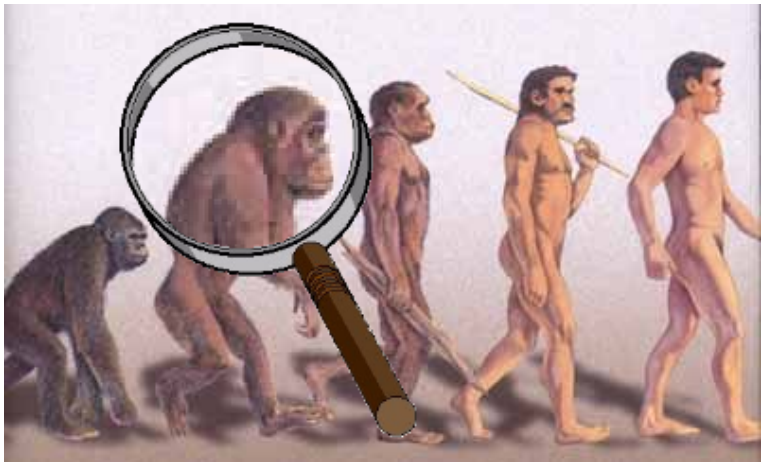
Lecture 12: *Stochastic Differential Equations*



Fisher-Wright model, SDEs, Fokker-Planck.

Outline

1. **Recap**
2. Diffusion Models
3. Fokker-Planck Equation
4. Steady State Distribution



Recap

- Recall the Fisher-Wright model looks at how a mutant allele at a single locus would take over the population
- For finite populations this is a stochastic process
- We can simulate this happening (each simulation will give us a different result)
- We can also compute $\boldsymbol{p}(t) = \big(p_0(t), p_1(t), \ldots, p_P(t)\big)$ telling us the probability of n mutants at time t using a Markov chain analysis
- Unfortunately, we can only do this for small populations

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Stochasticity in Biology

- By its very nature dynamics in biology is very random
- Sometimes this doesn't matter (we can ignore the randomness)
- But sometimes this randomness is an essential part of nature
- This is particularly true when the dynamics depends on rare events that might not happen
- This is true in population genetics, but also in gene regulation which we explore later in the course

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Language of Random Dynamics

- When the underlying dynamics involve small random kicks then we can develop a calculus known as **stochastic differential equation**
- This is used across the sciences
- It was originally used to describe physical diffusion
- It is used, for example, in describing fluctuations in finance

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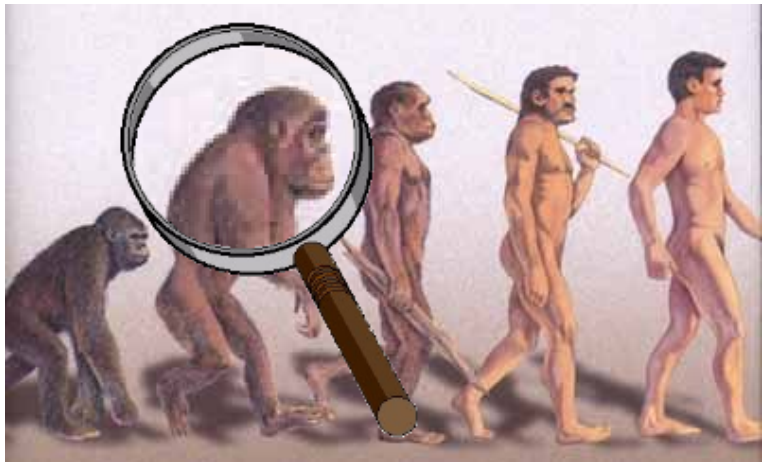

Diffusion Models

- Markov analysis quickly becomes intractable
- For weak selection and small mutation rates we can model the system using a *stochastic difference equation*
- Let $X(t)$ be the proportion of mutants at time t
- We make the assumption that at time $t + \Delta t$ the proportion of mutants is given by

$$X(t + \Delta t) = X(t) + a(X(t))\Delta t + b(X(t))\Delta W(t)$$

- where $\Delta W(t)$ is a random variable with mean 0 and variance Δt

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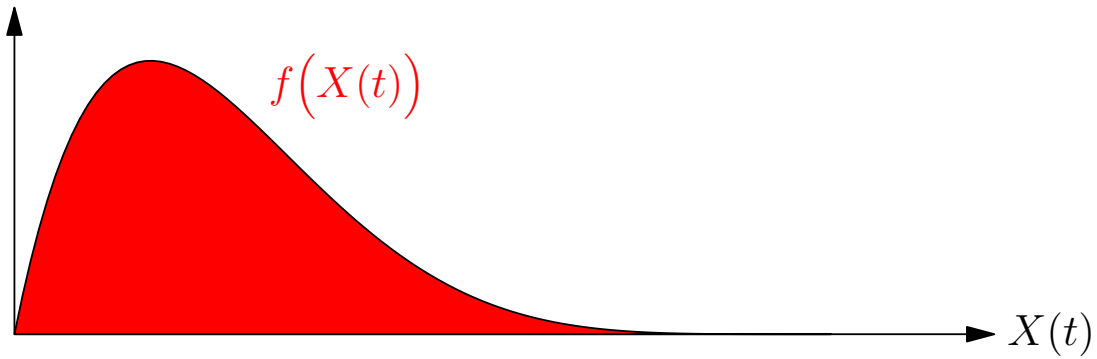
Random Variables

- $X(t)$ (the proportion of mutants at time t) is a random variable
- That is, it varies from realisation to realisation
- $X(t)$ will be distributed according to a probability density $f(X(t))$



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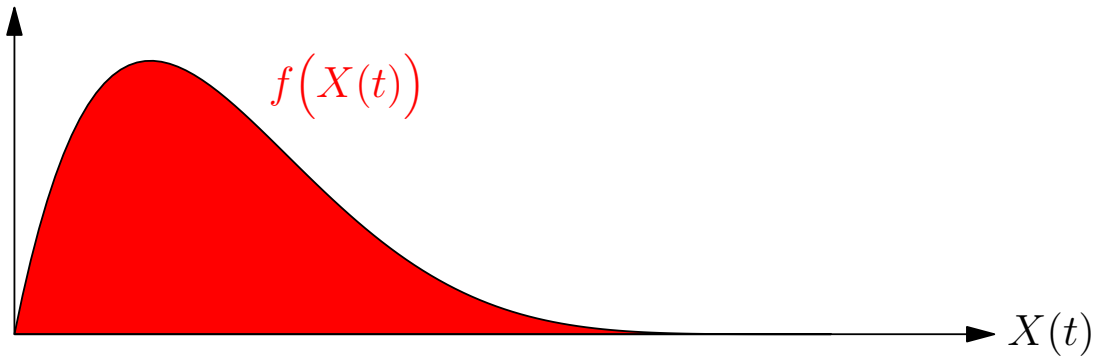

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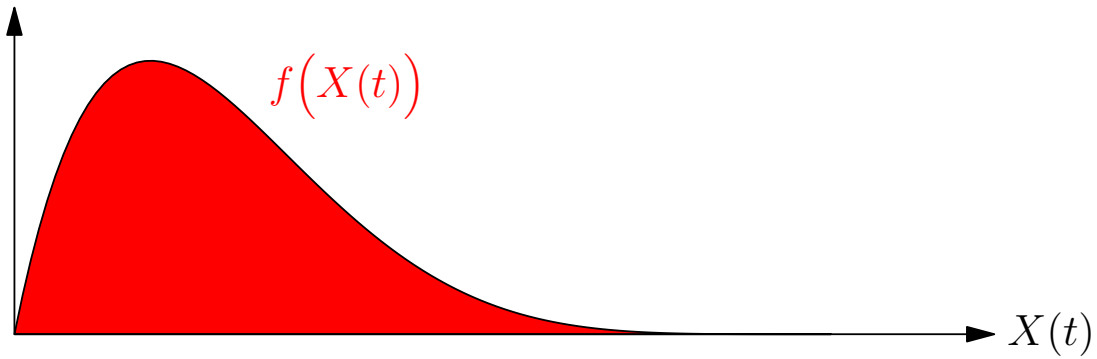

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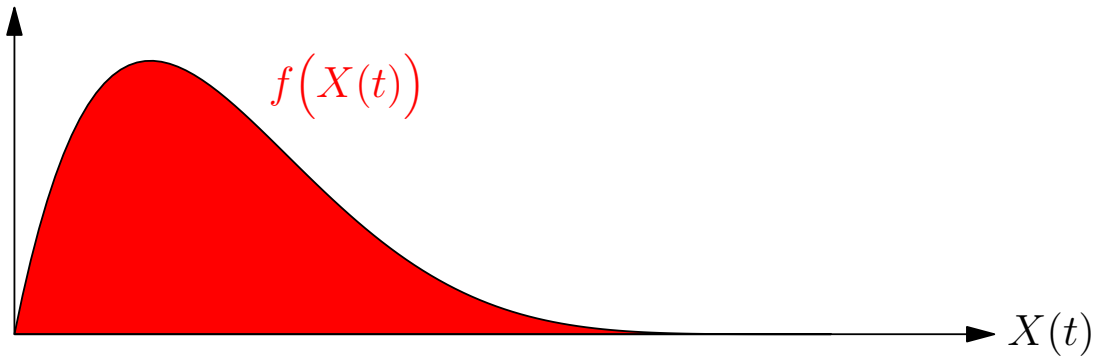

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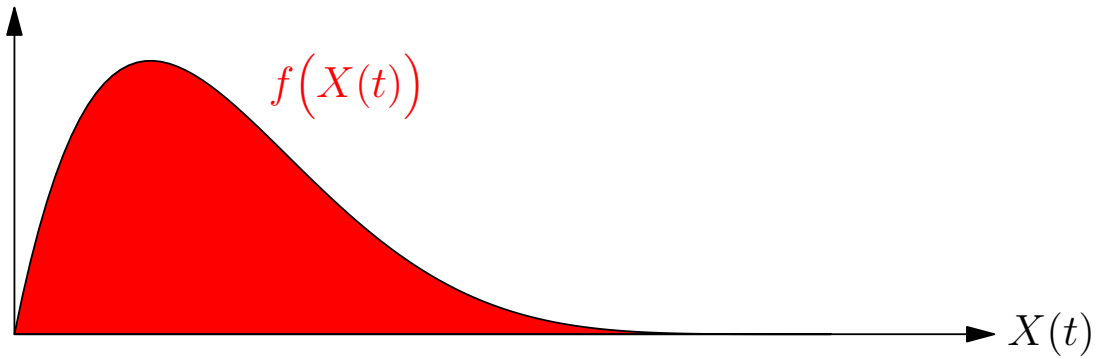

- We denote its average or **expectation** as

$$\mathbb{E}\left[X(t)\right] = \int X(t)\, f(X(t))\, \mathrm{d}t = \langle X(t) \rangle$$

Naming and Mis-naming

- On average

$$\mathbb{E}\left[X(t+\Delta t)\right] = X(t) + a(X(t))\Delta t$$

  this defines the function $a(X(t))$

- Physicists call this the *drift term*

- The variance in $X(t+\Delta t)$ is given by

$$\mathbb{E}\left[X^2(t+\Delta t)\right] - \mathbb{E}\left[X(t+\Delta t)\right]^2 = b^2(X(t))\Delta t$$

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Fisher-Wright Average

- For the Fisher-Wright model we have

$$\mathbb{P}\left(X(t+\Delta t)=x\right)=\mathbb{P}\left(n'=Px\right)\;=\binom{P}{Px}p_{sm}^{Px}(1-p_{sm})^{P(1-x)}$$

- For this model it makes sense to take $\Delta t = 1$ (one generation)

- Taking averages

$$\begin{aligned} a(X(t)) &= \mathbb{E}\left[X(t+1)\right]-X(t)=p_{sm}-X(t) \\ &= \frac{\left(u-(u+v-s+u\,s)X(t)+sX^2(t)\right)}{1+s\,X(t)} \end{aligned}$$

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Fisher-Wright Variance

- The variance of a binomial is given by

$$b^2(X(t)) = \mathbb{E}\left[X^2(t+1)\right] - \mathbb{E}\left[X(t+1)\right]^2 = \frac{p_{sm}(1-p_{sm})}{P}$$

- If s, u and v are small then $p_{sm} \approx X(t)$ so

$$b^2(X(t)) \approx \frac{X(t)\left(1-X(t)\right)}{P}$$

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Continuous Time

- Taking the limit $\Delta t \to 0$ we obtain a (Itô) *stochastic differential equation* (SDE)

$$\mathrm{d}X(t) = a(X(t))\,\mathrm{d}t + b(X(t))\,\mathrm{d}W(t)$$

- This has to be viewed as a formal definition—its precise meaning is rather technical

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Why SDE Work

- SDE's model fluctuating systems using two terms
  - ⋆ The mean change described by $a(X(t))\,\mathrm{d}t$ which we can model accurately (it become more accurate in large populations where the difference between discrete and continuous time is negligible)
  - ⋆ Fluctuations $b(X(t))\,\mathrm{d}W(t)$ which are assumed to be normally distributed

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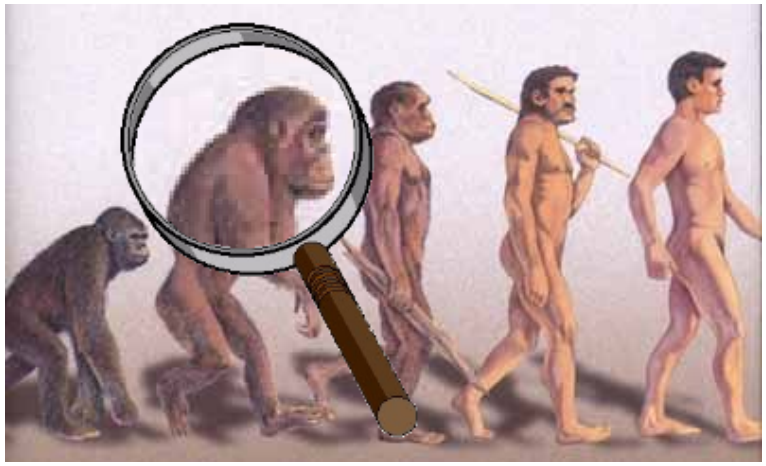

Random Instances and Distributions

- The stochastic differential equation (aka diffusion equation) describes $X(t)$ in terms of a second stochastic process $W(t)$
- $W(t)$ is known as a Weiner measure it has the distribution

$$f\big(W(\Delta t + t_0) - W(t_0) = x\big) = \mathcal{N}(x|0, \Delta t)$$

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- The Fokker-Planck equation helps in that rather than work with the weird world of stochastic differential equation we can work with the well understood partial differential equations (PDEs)
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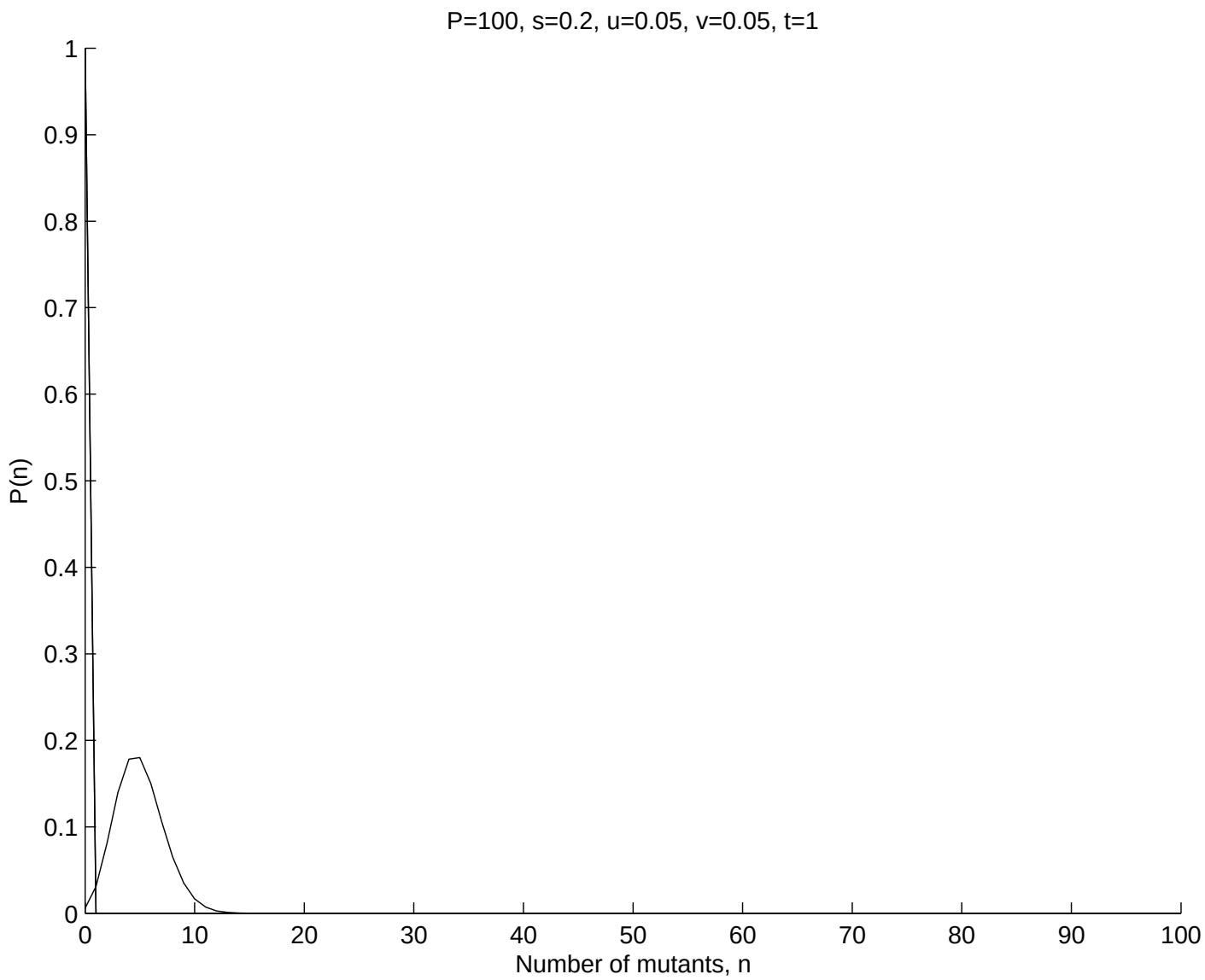

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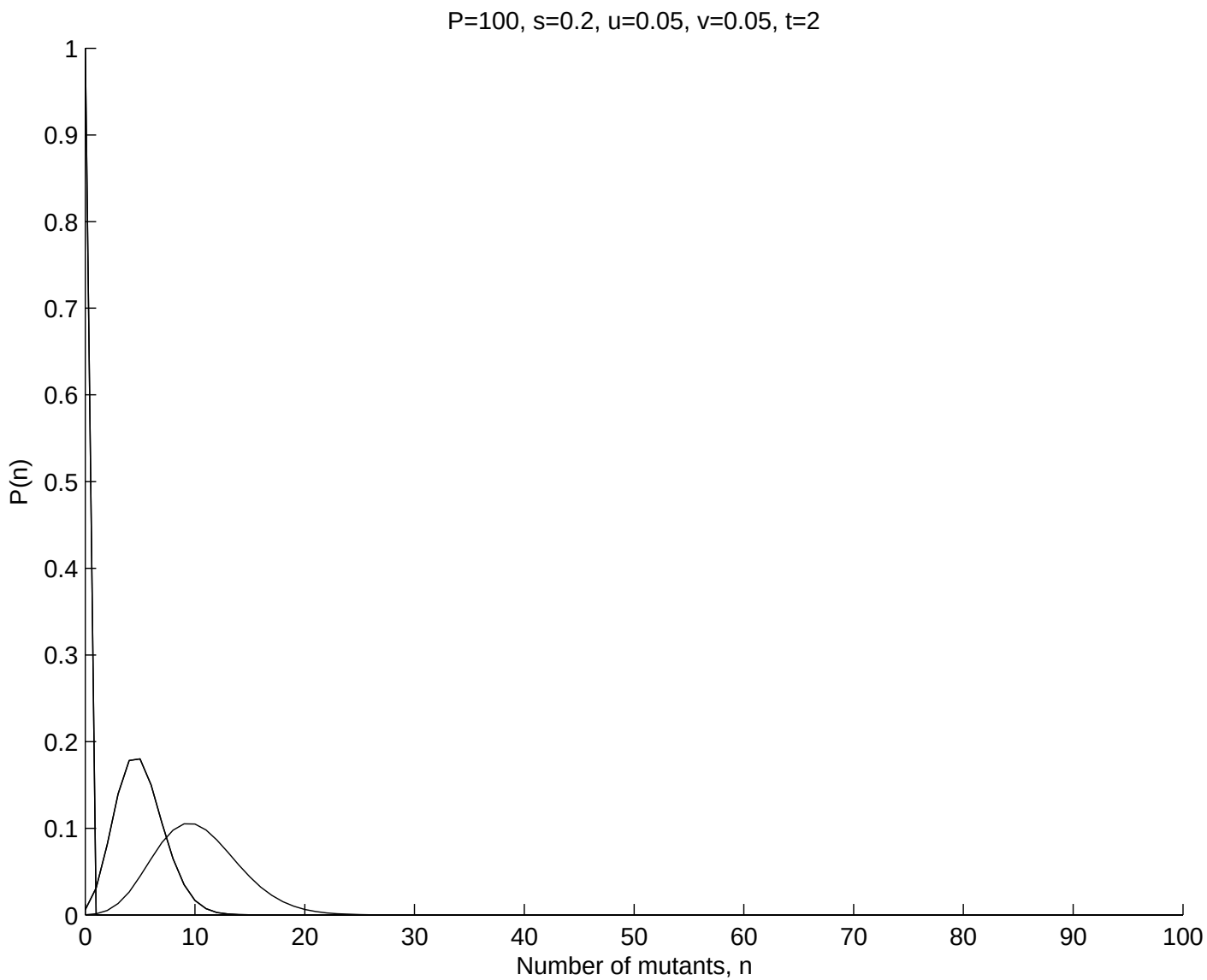

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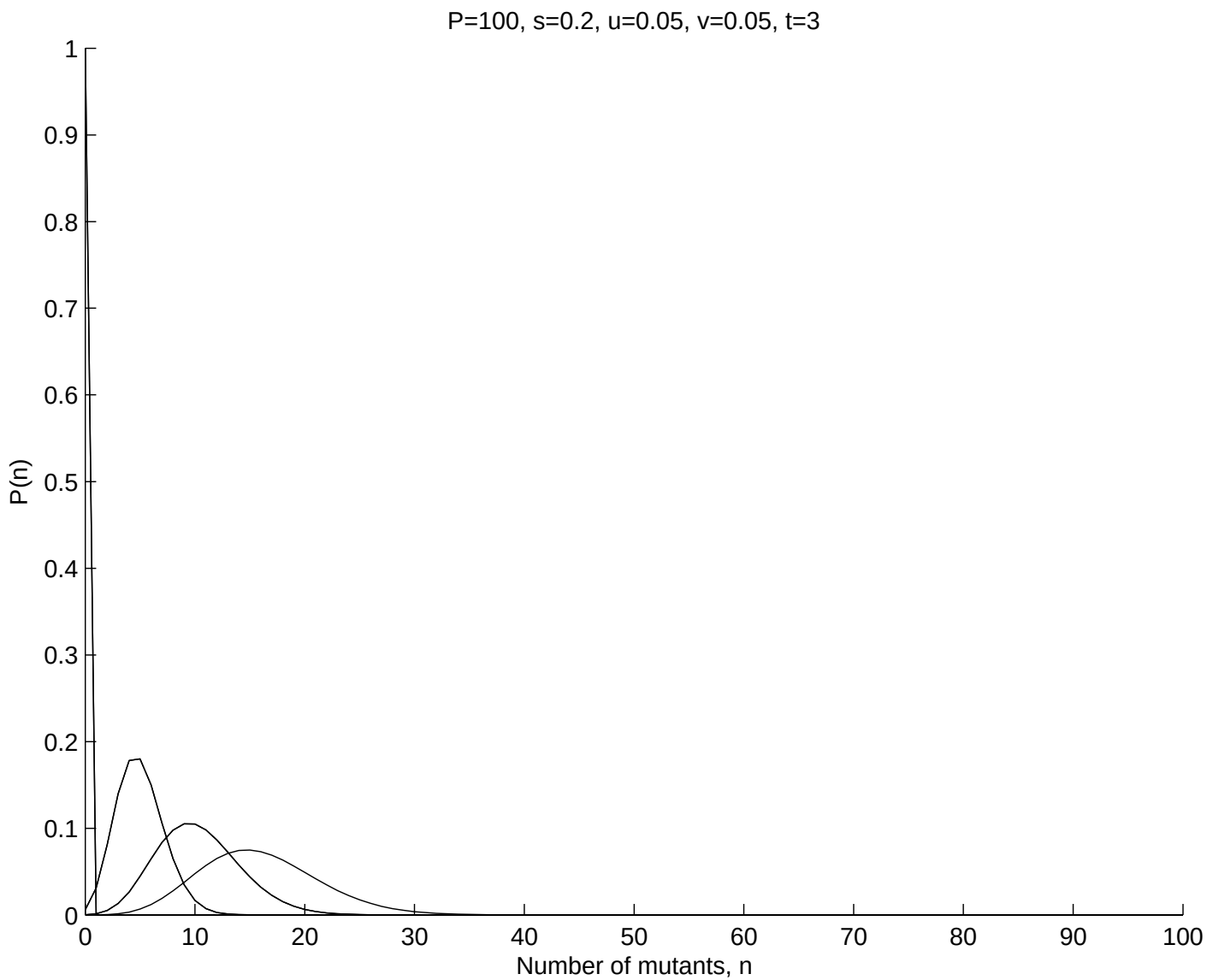

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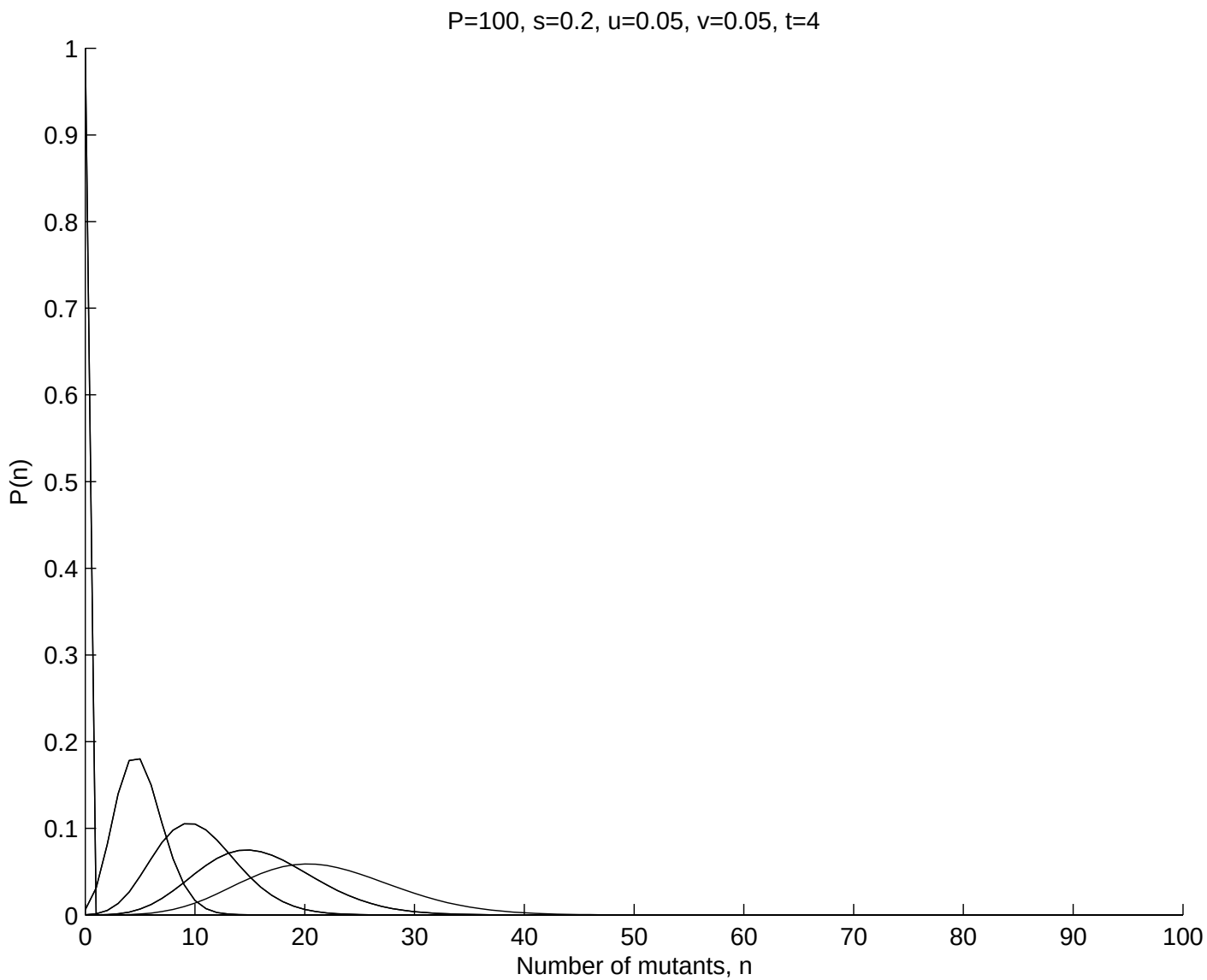

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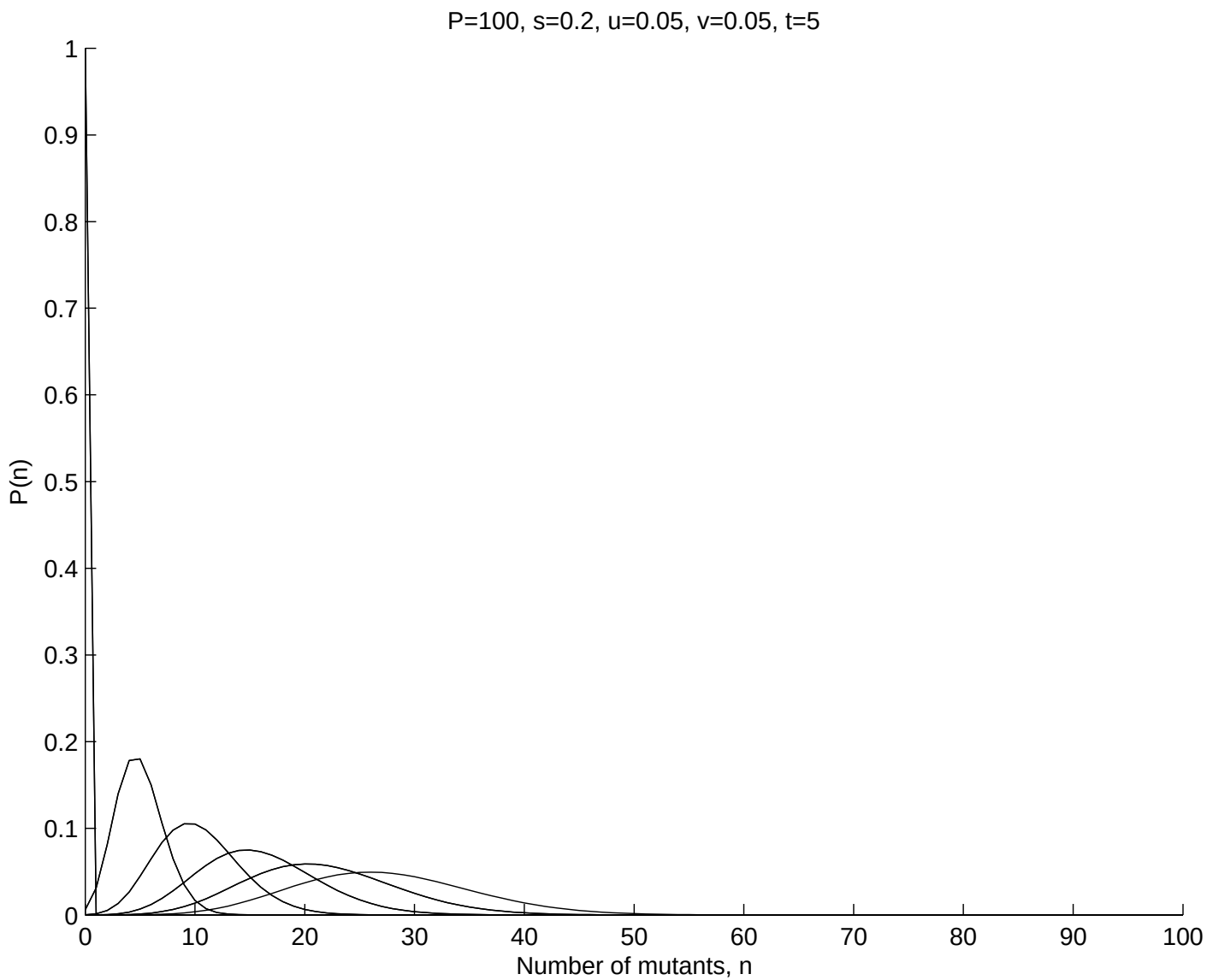

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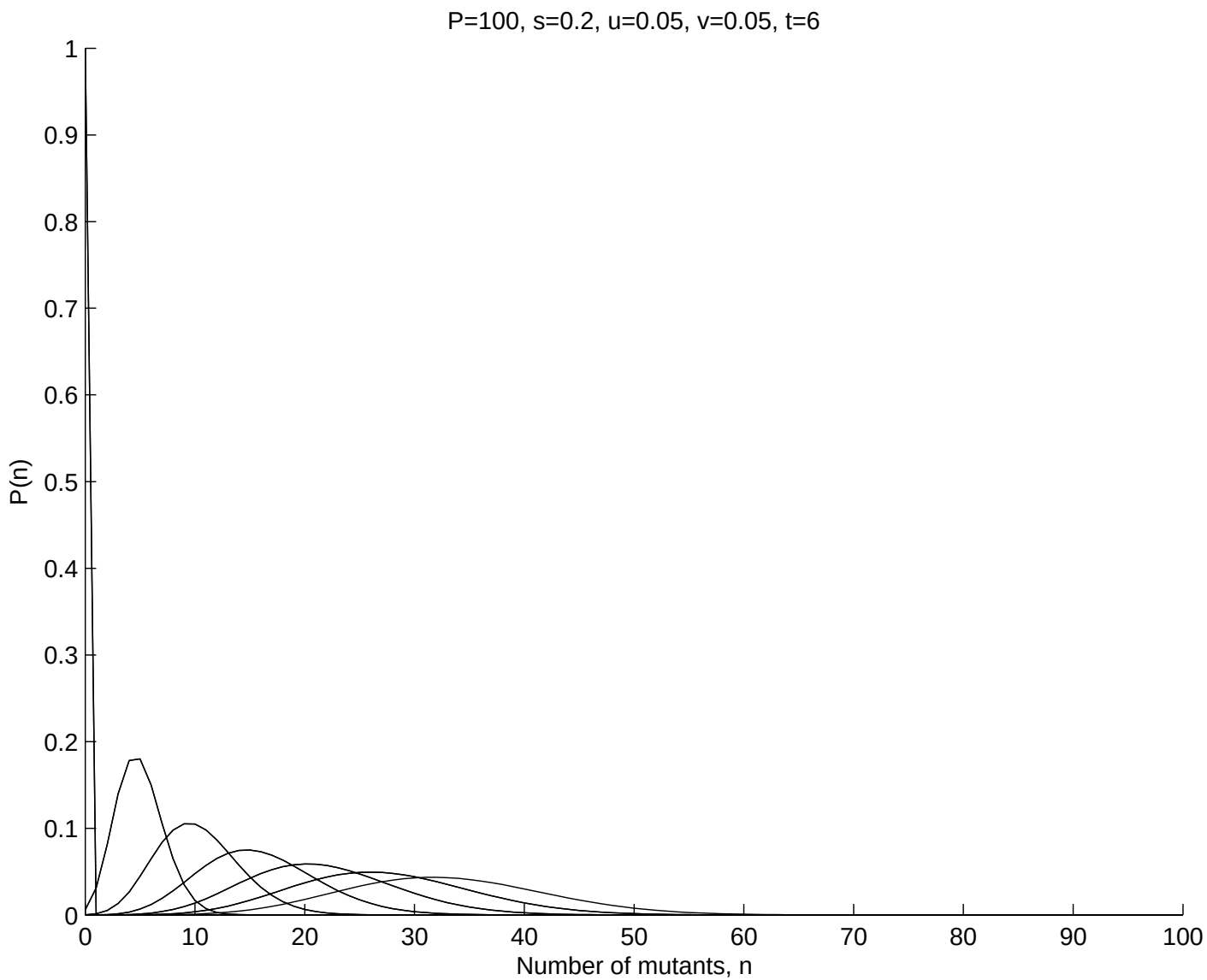

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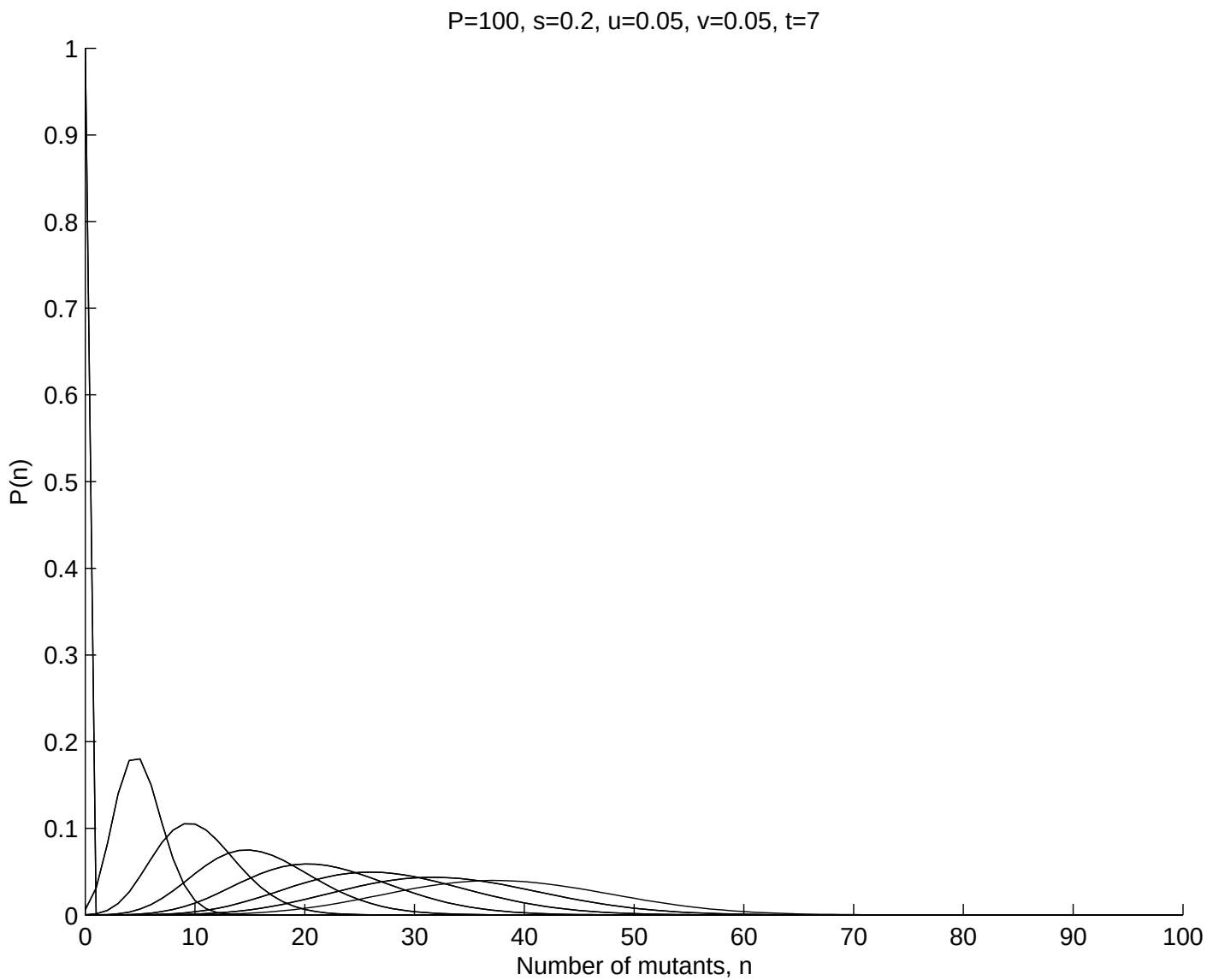

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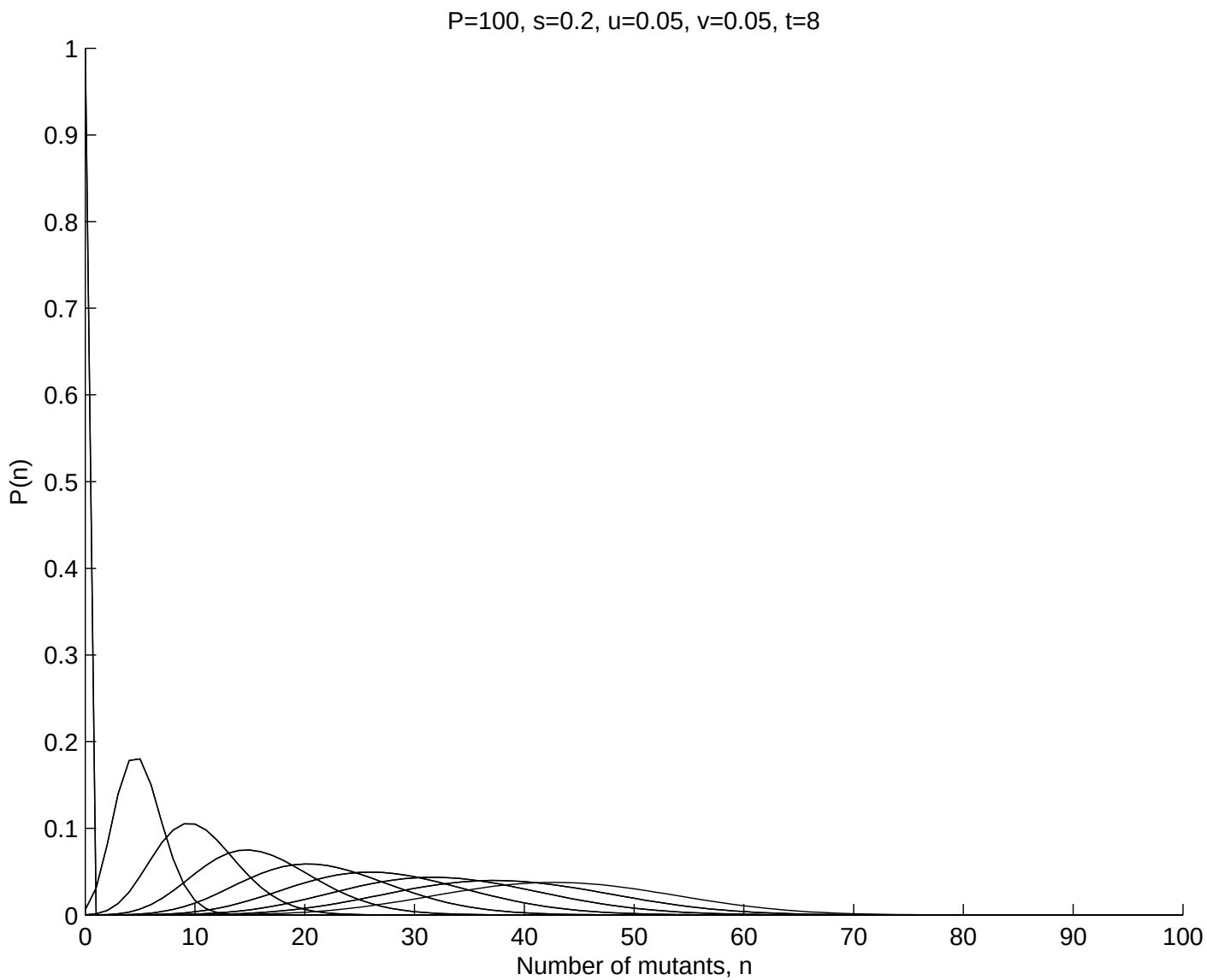

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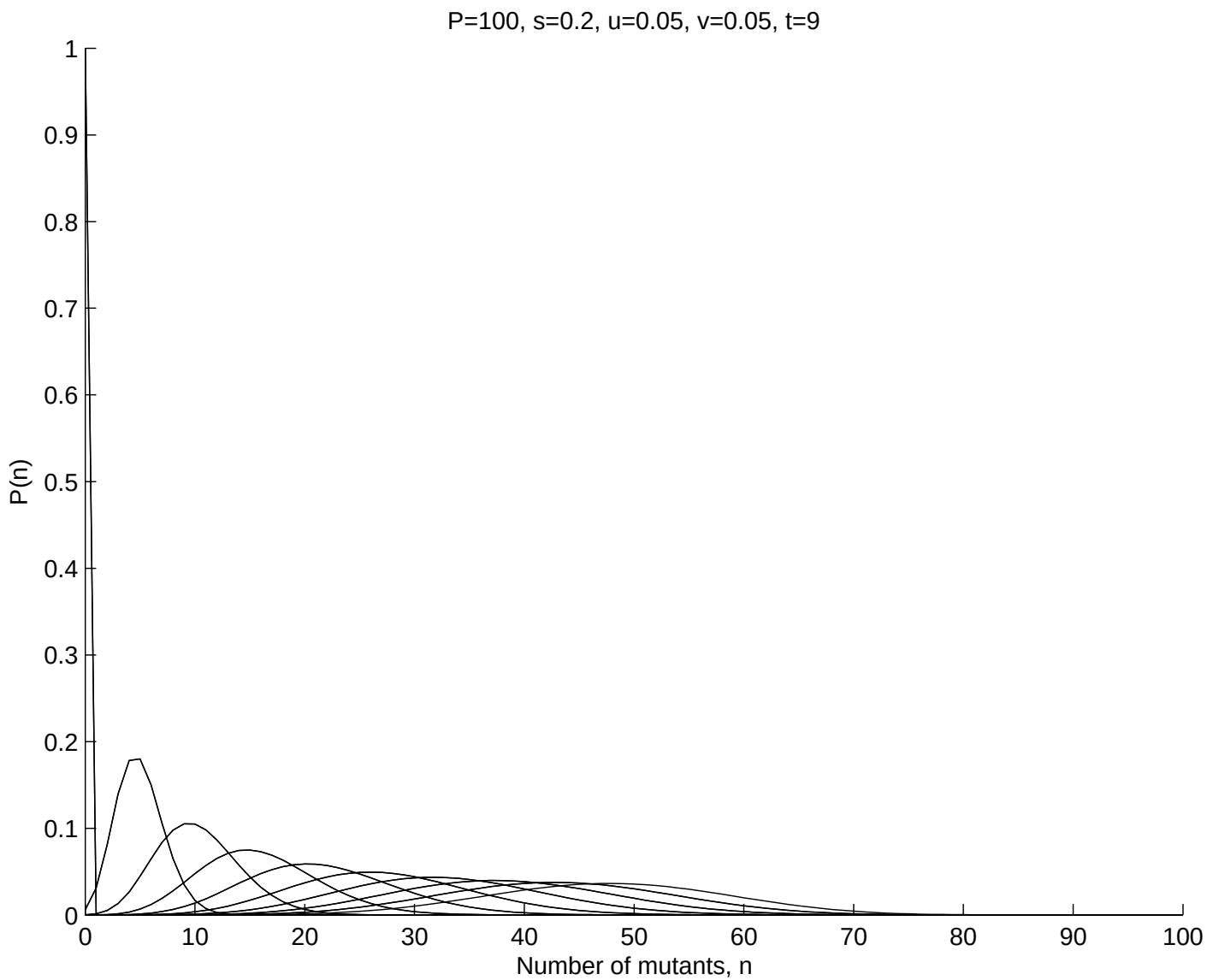

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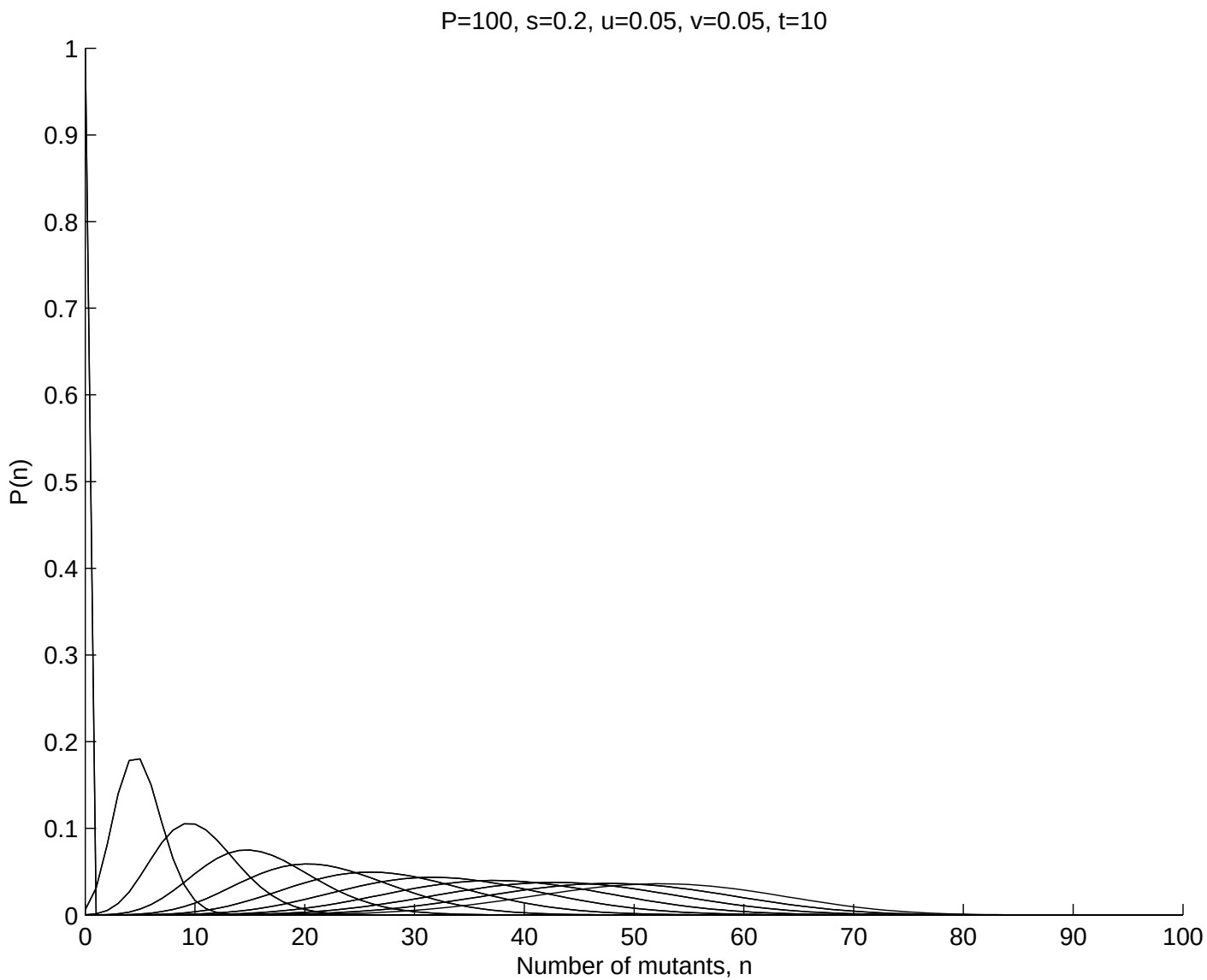

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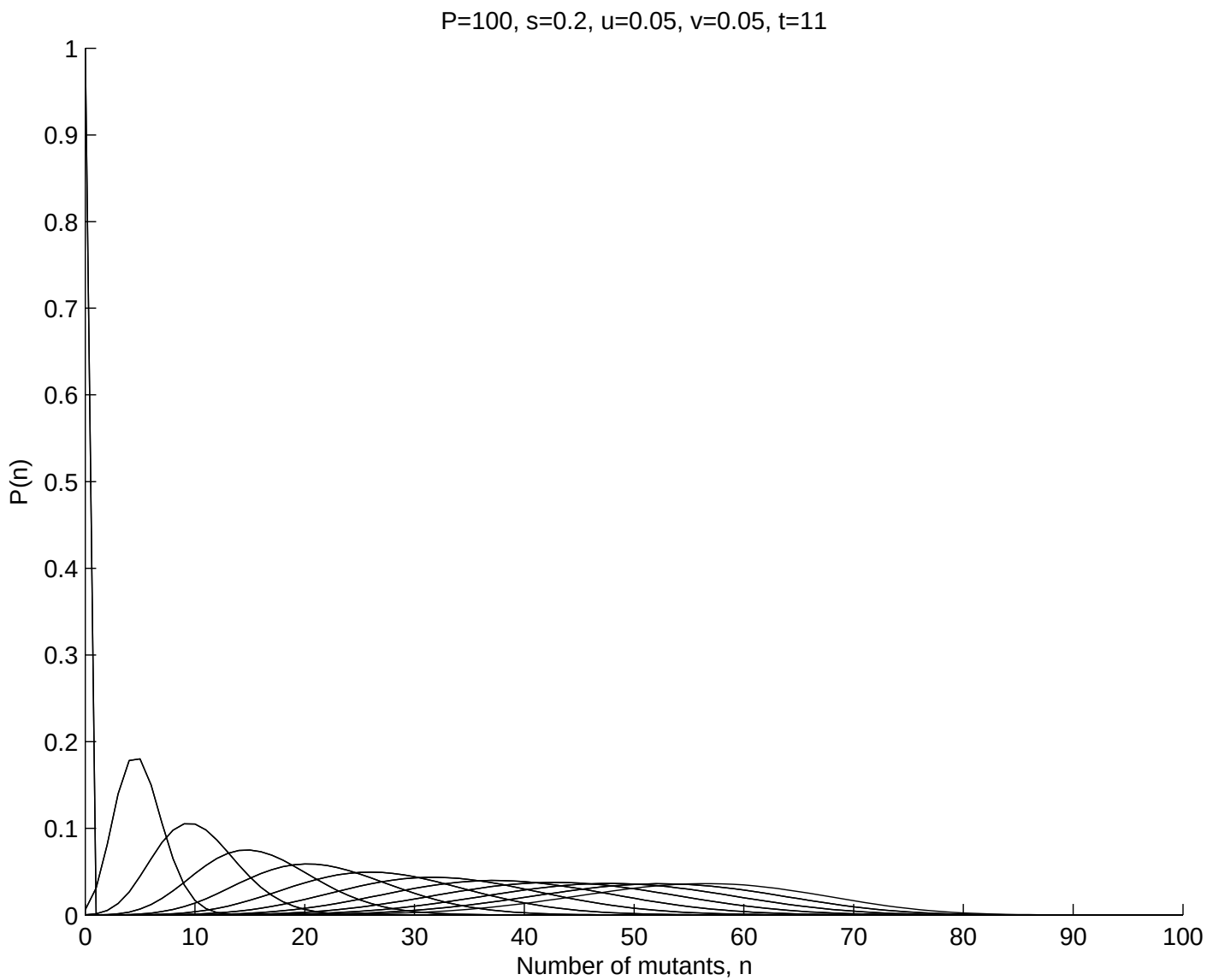

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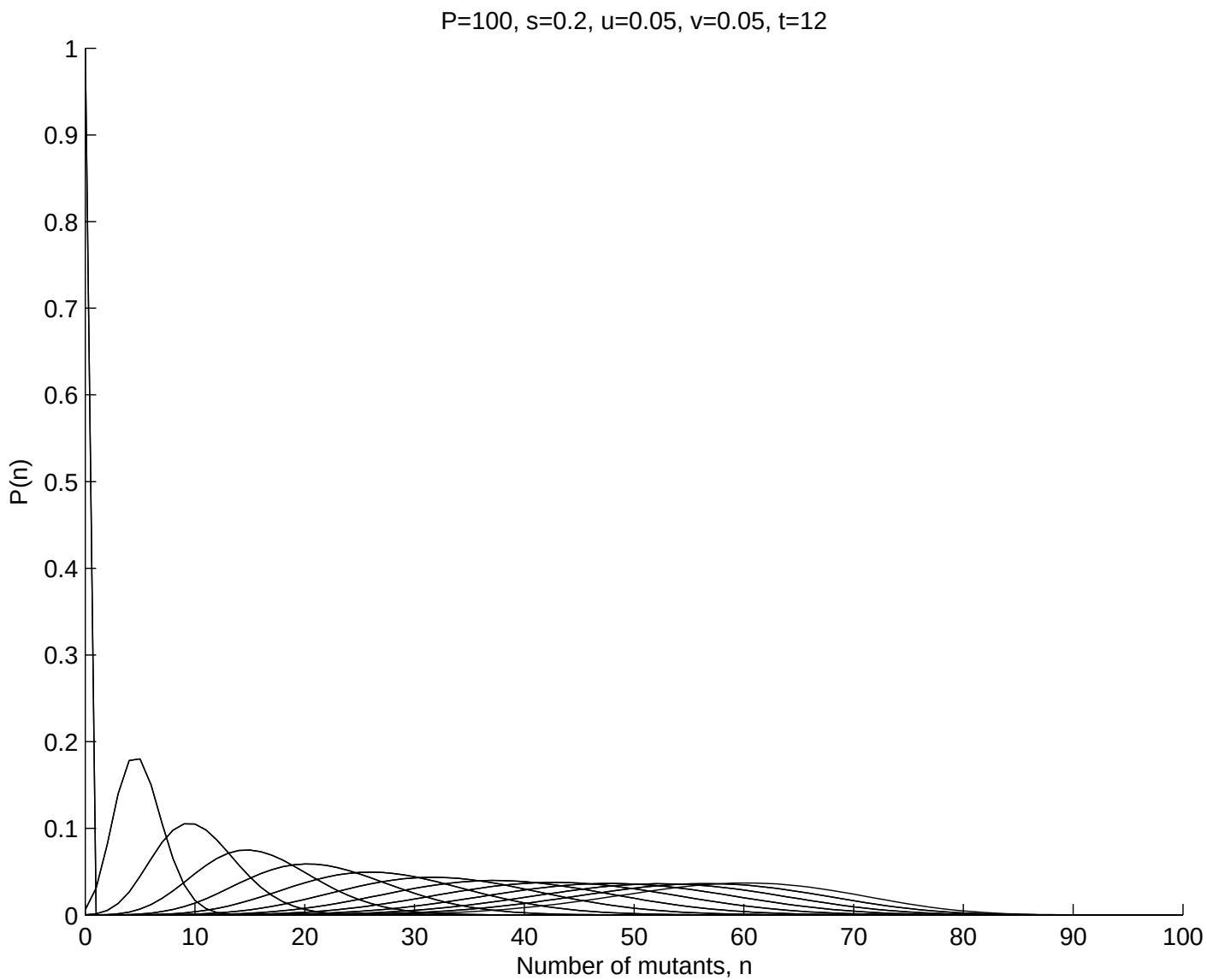

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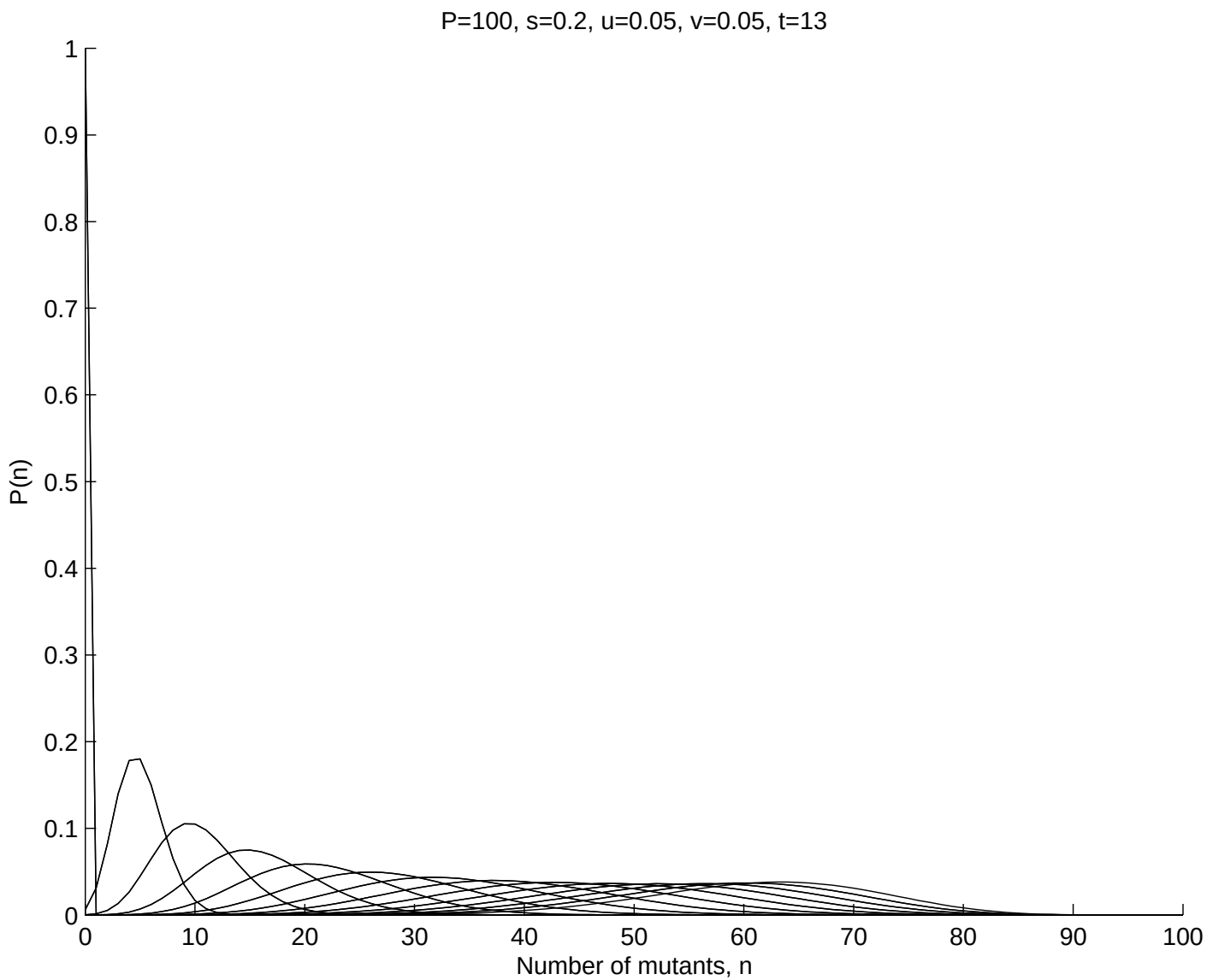

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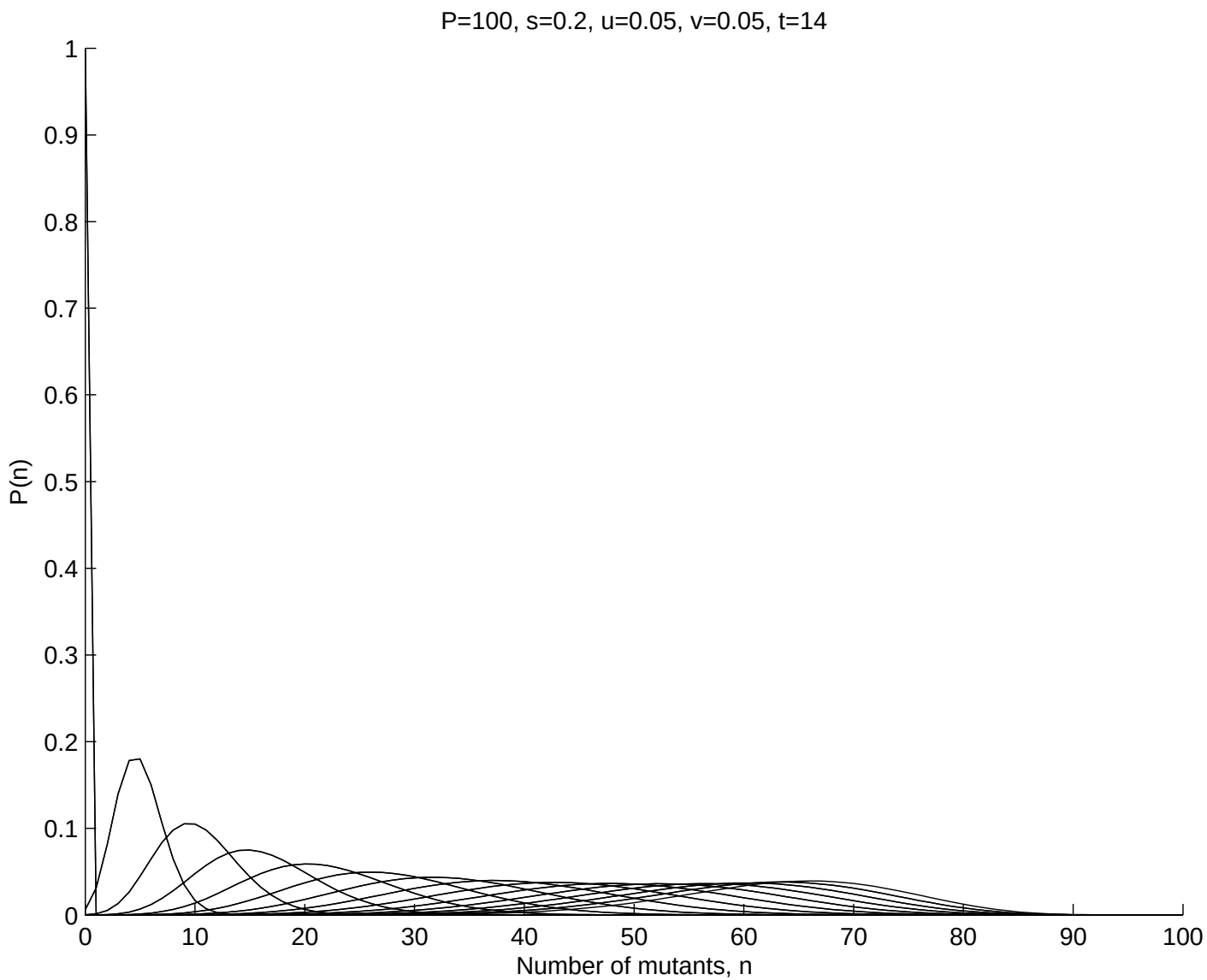

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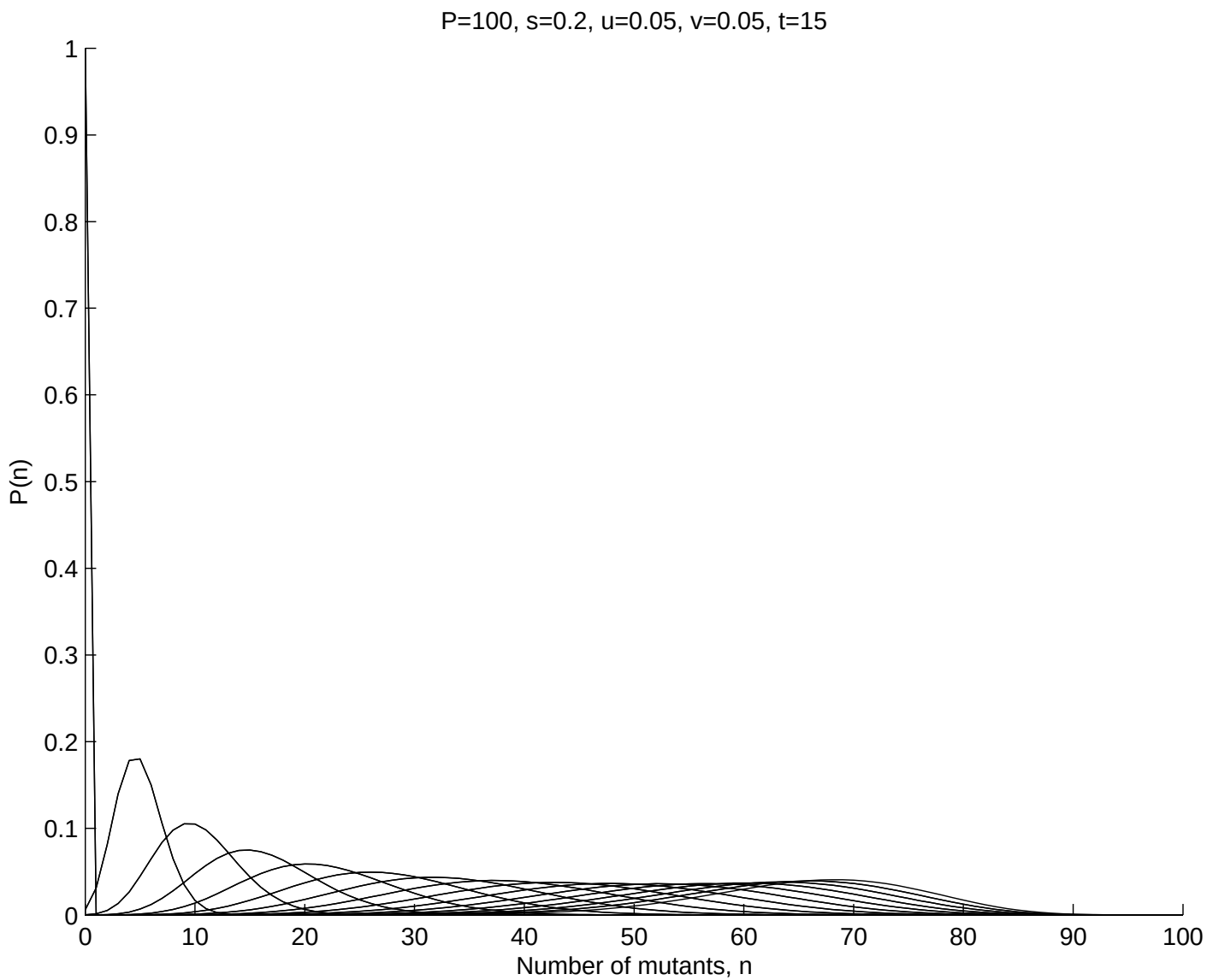

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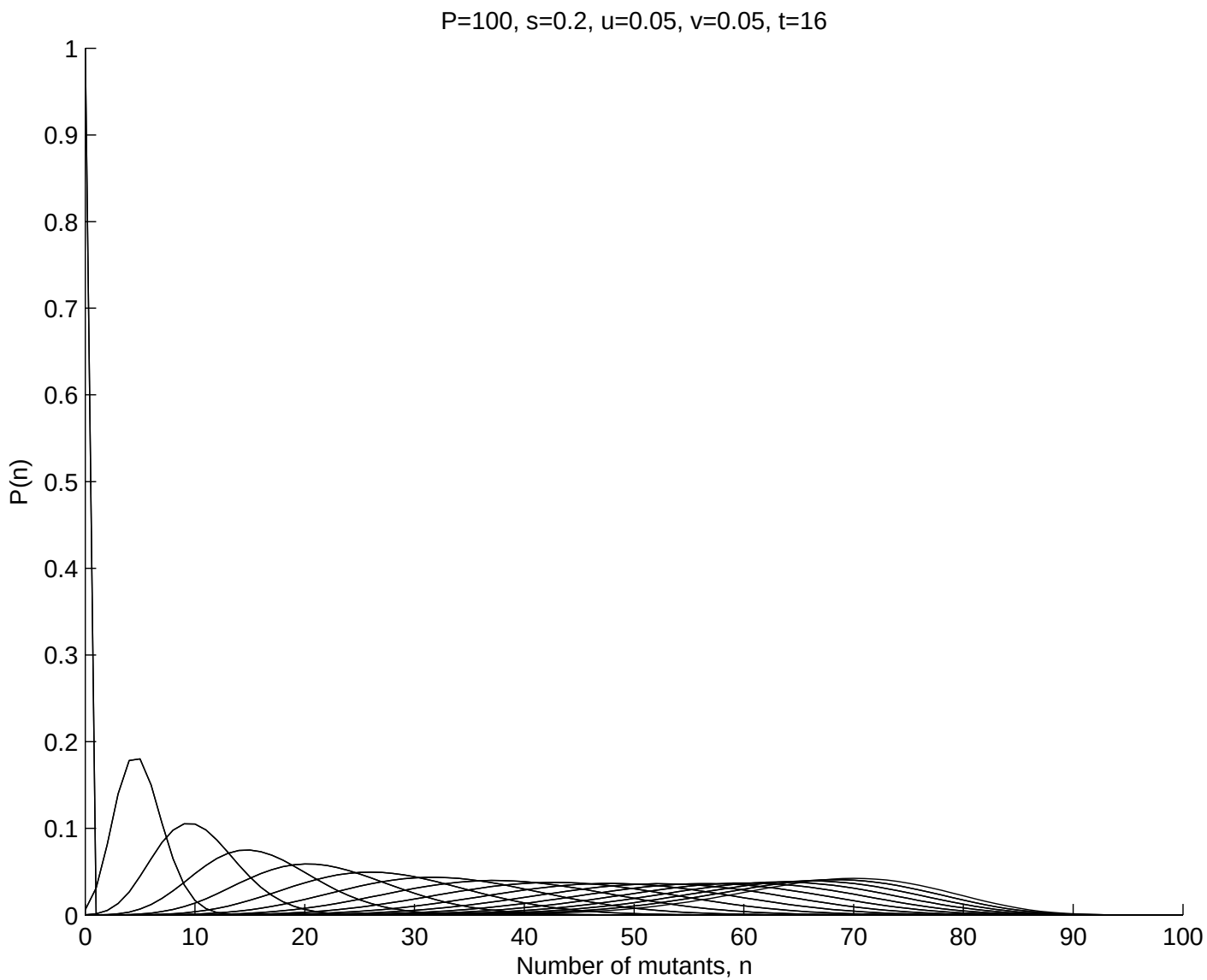

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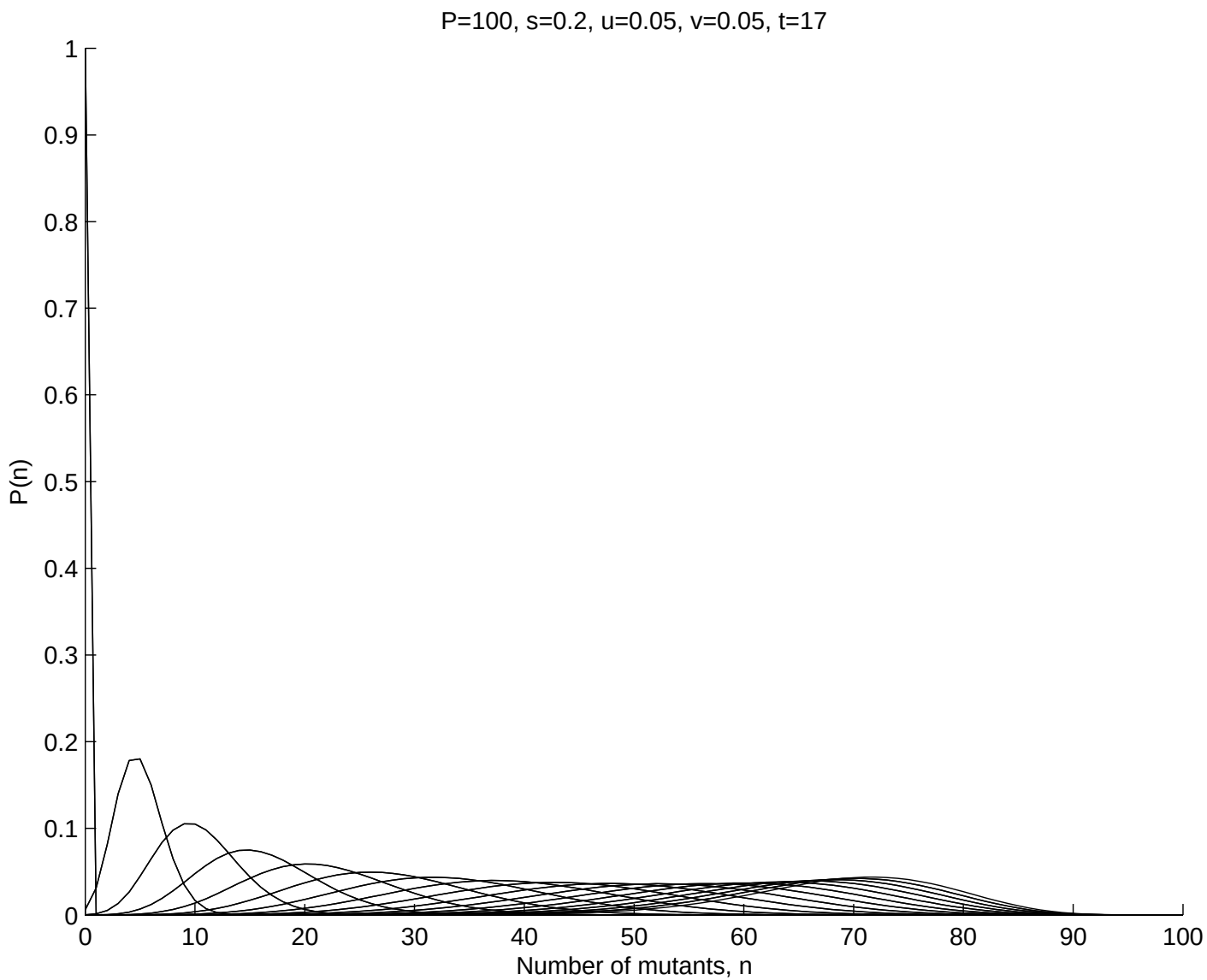

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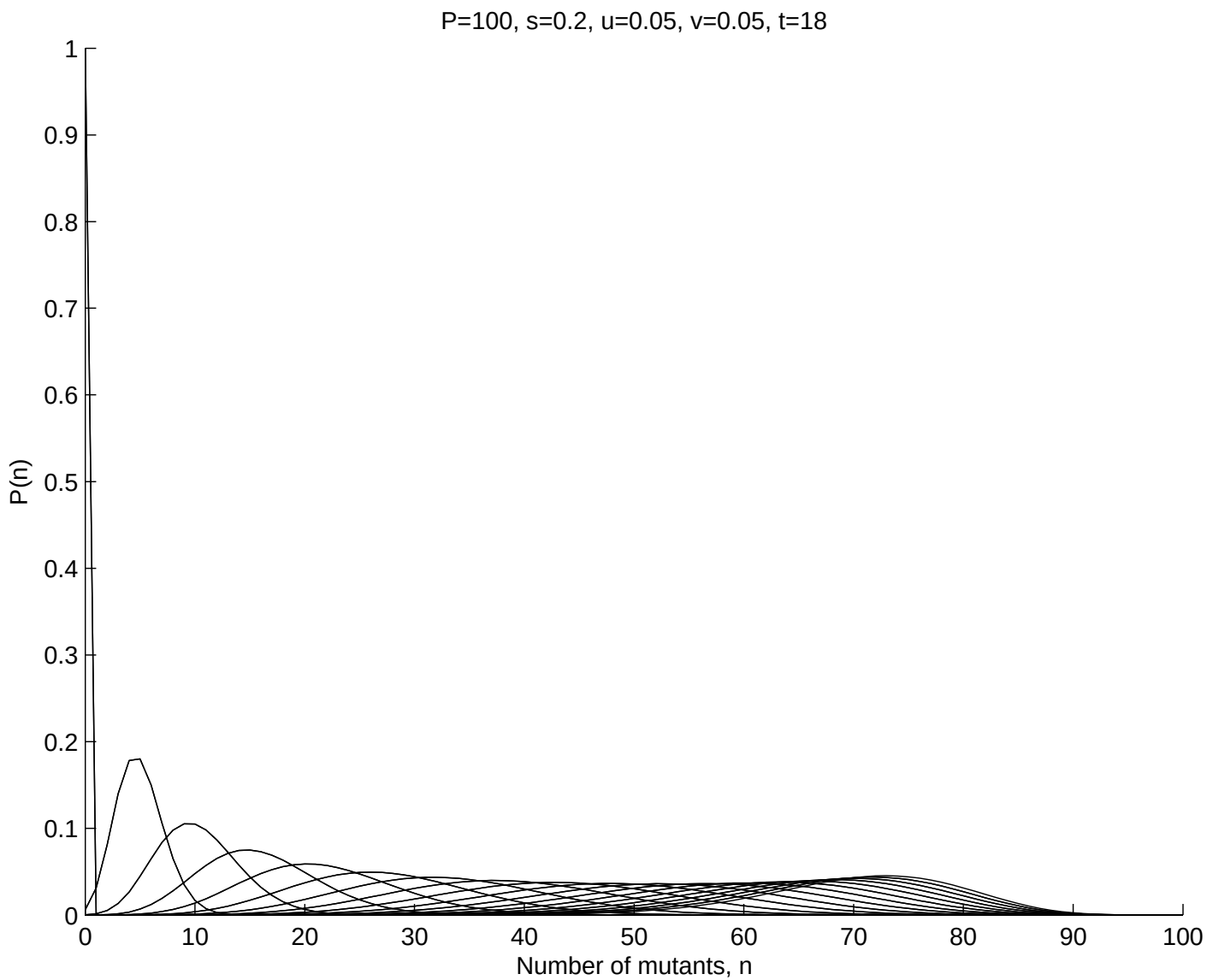

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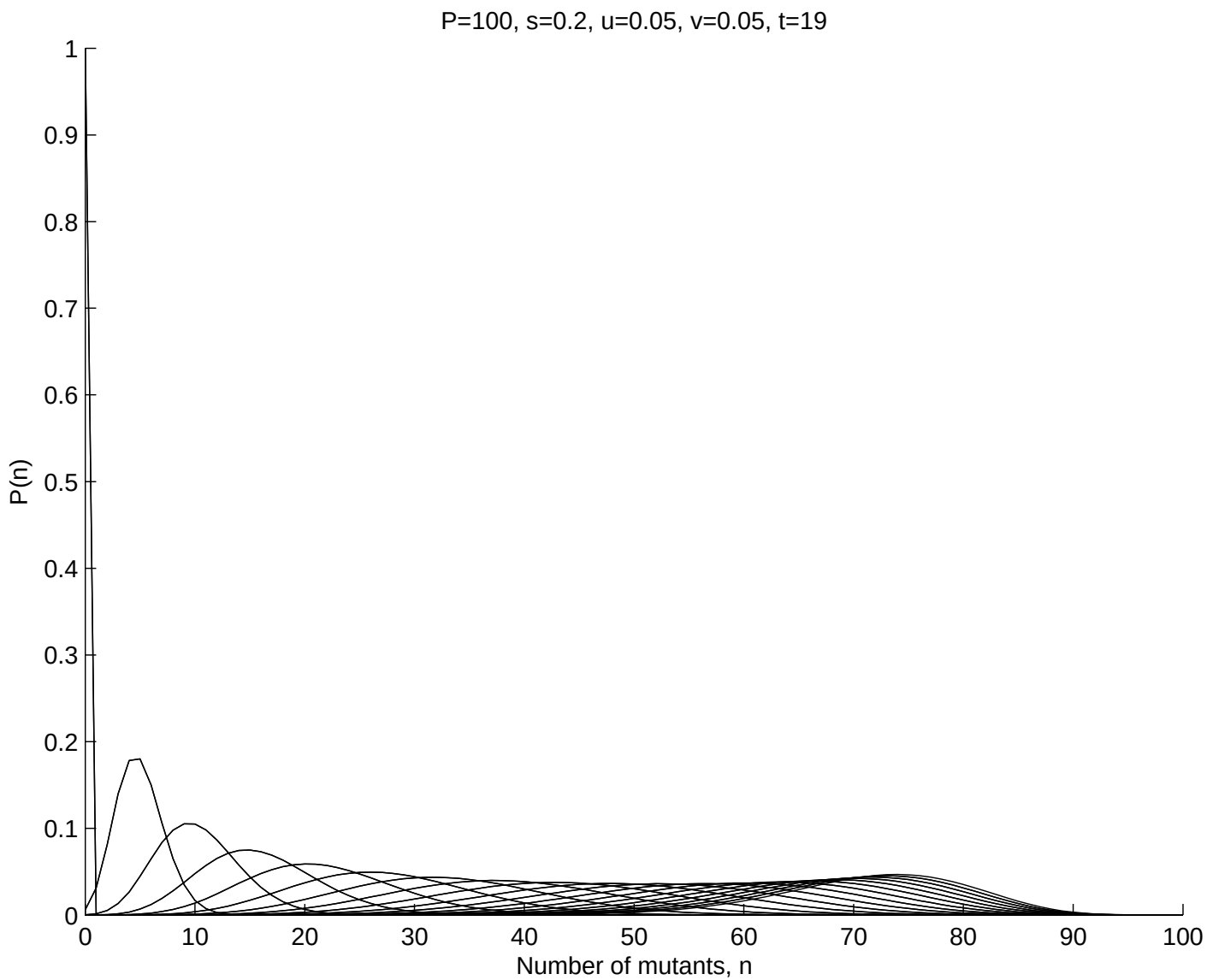

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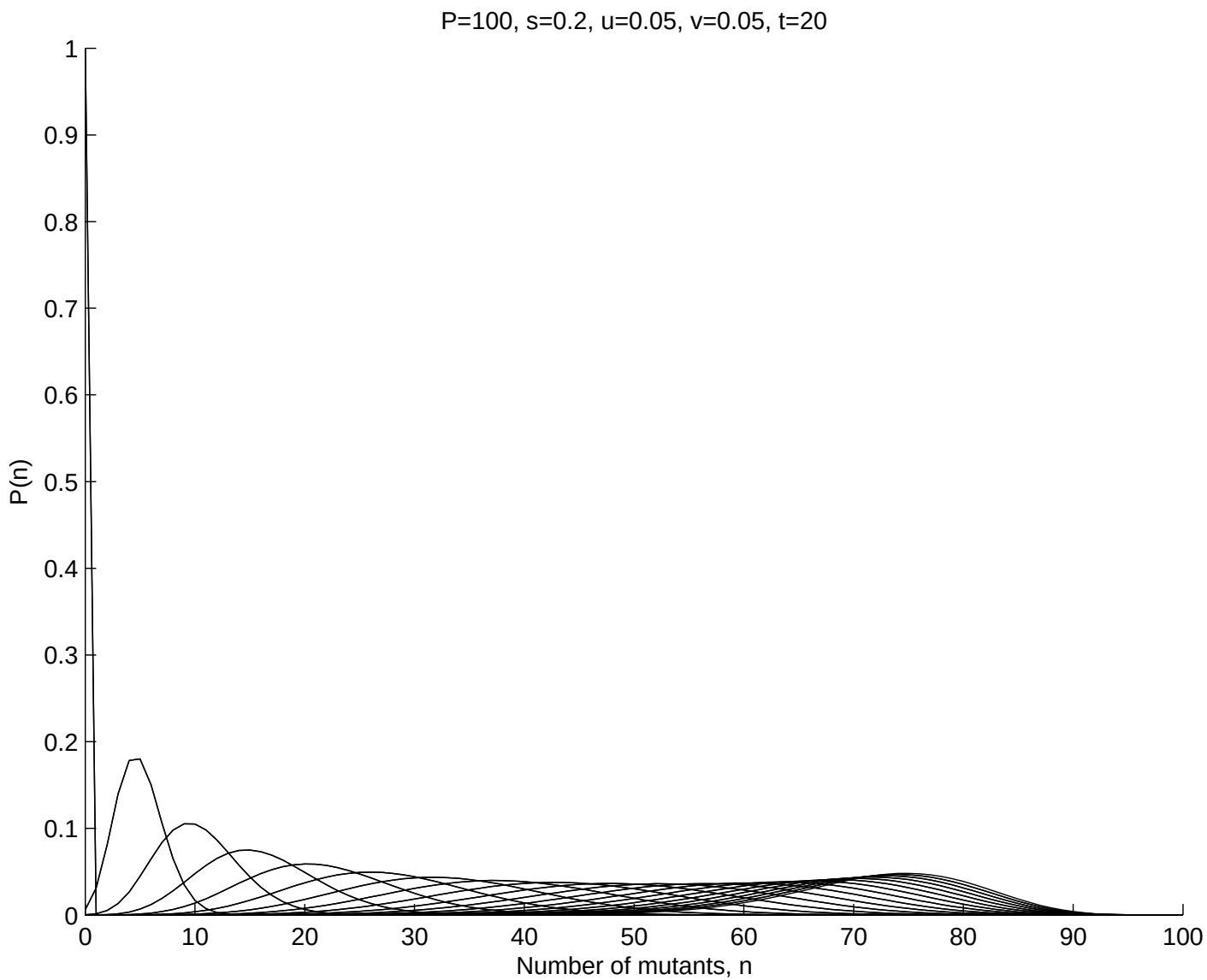

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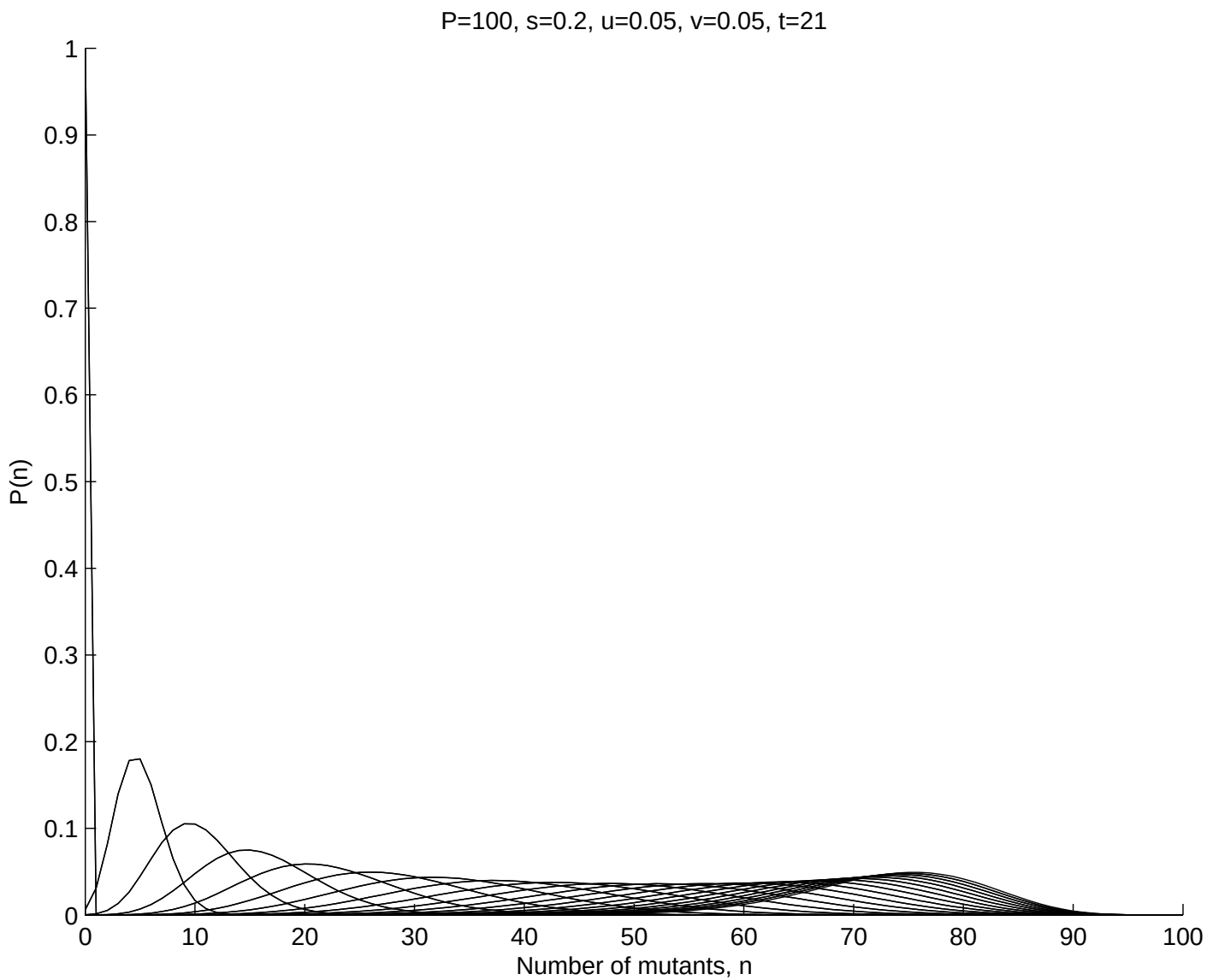

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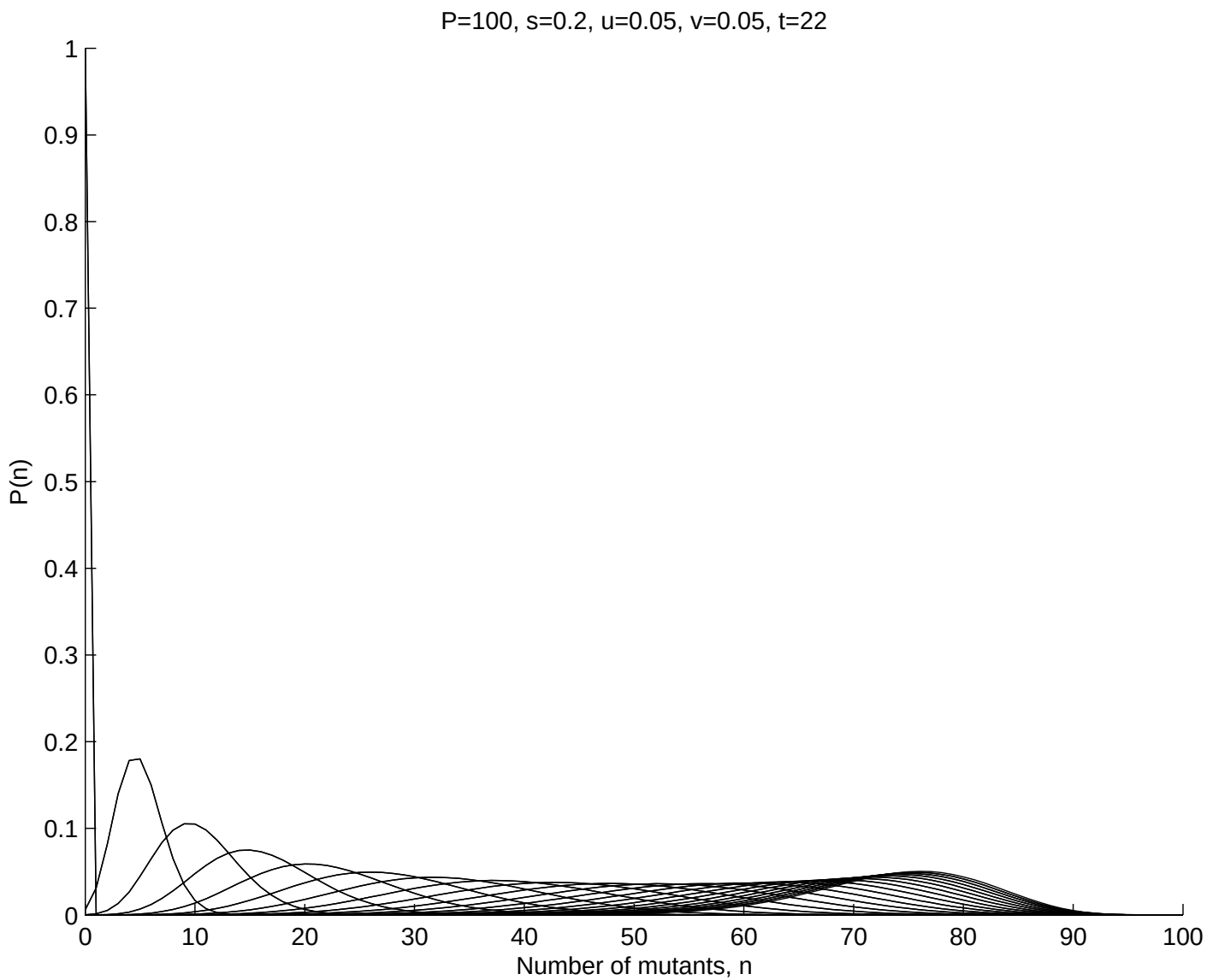

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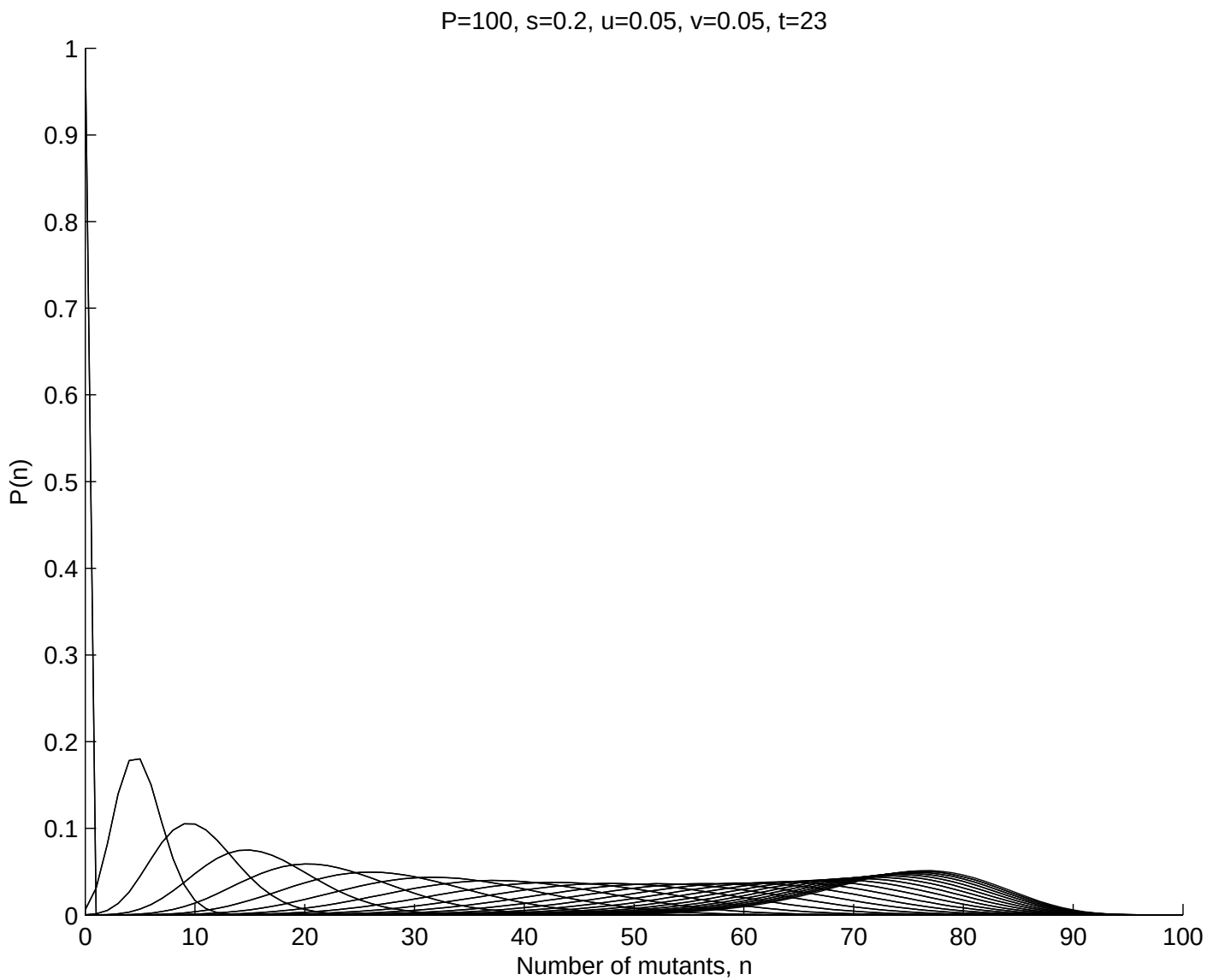

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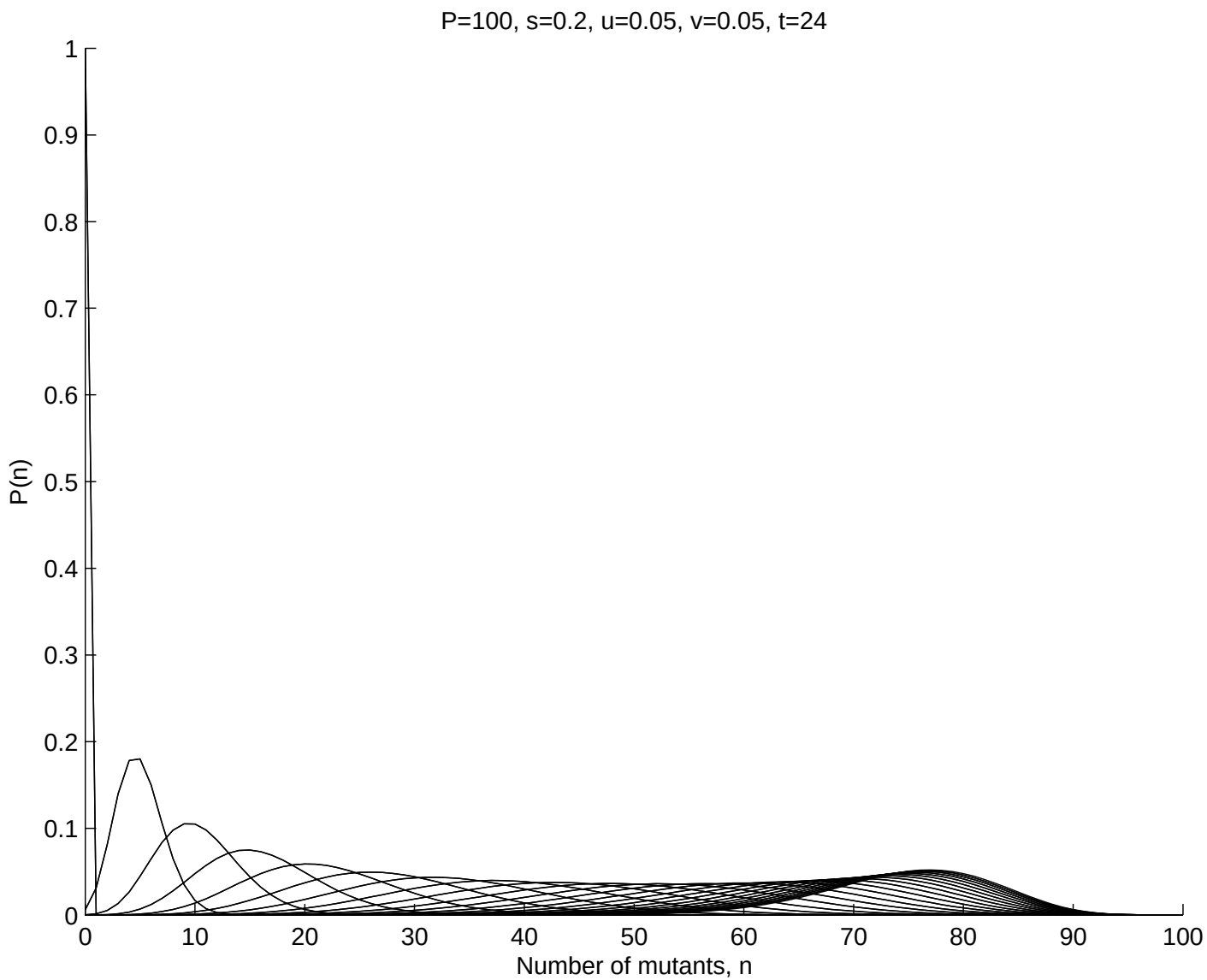

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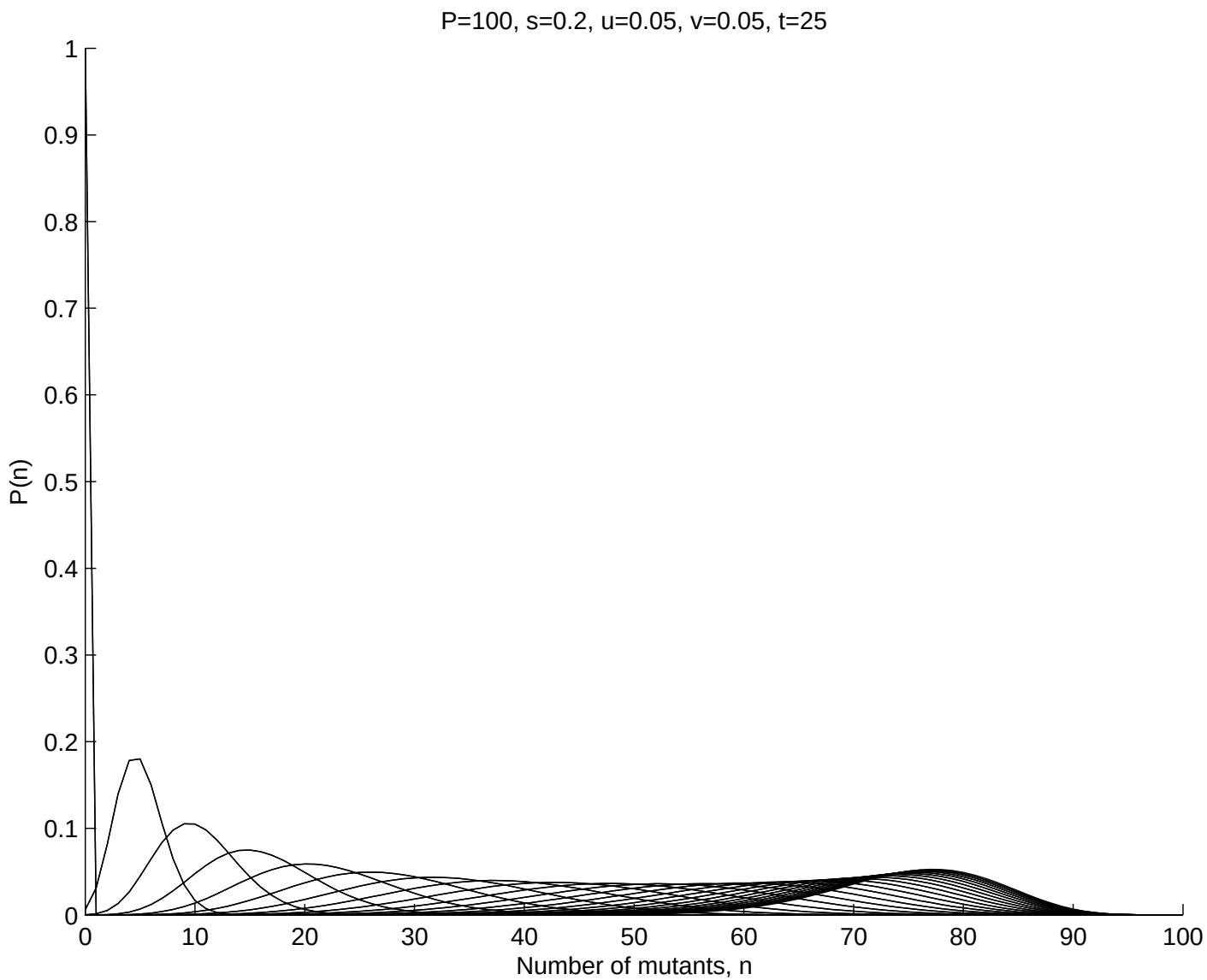

Solution to Fokker-Planck

- I cheated, this is actually the solution to the Markov chain
- However, the Fokker-Planck provides the same information and would provide an almost identical outcome
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Outline

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2. Diffusion Models
3. Fokker-Planck Equation
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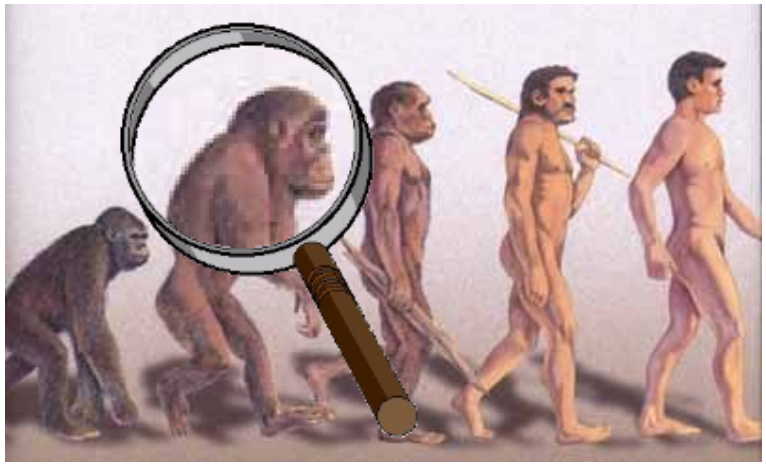

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- The steady state solution $f_{eq}(x) = \lim_{t\to\infty} f(x,t)$ does not depend on t so the Fokker-Planck equation becomes an ordinary differential equation

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Steady State for Fisher-Wright

- Assuming s, u and v are small, then expanding to first order

$$a(X) = \mathbb{E}\left[\Delta X\right]/\Delta t \approx -v\,X + u(1-X) + sX(1-X)$$

$$b(X) = \left(\mathbb{E}\left[(\Delta X)^2\right] - \mathbb{E}\left[\Delta X\right]^2\right)/\Delta t \approx \frac{X(1-X)}{P}$$

- Using these values we find

$$f_{eq}(x) = \frac{x^{2\,P\,u-1}\,(1-x)^{2\,P\,v-1}\,\mathrm{e}^{2\,P\,s\,x}}{B(2\,P\,u, 2\,P\,v)\,F(2\,P\,u; 2\,P(u+v); 2\,P\,s)}$$

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What's the Point

- This seems a long journey to get a complicate formula, but this tells us the final distribution as a function of the parameters of our model
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Sickle-Cell Anaemia

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- This is caused by a particular mutation, but carriers only suffer if they have two copies of the mutation
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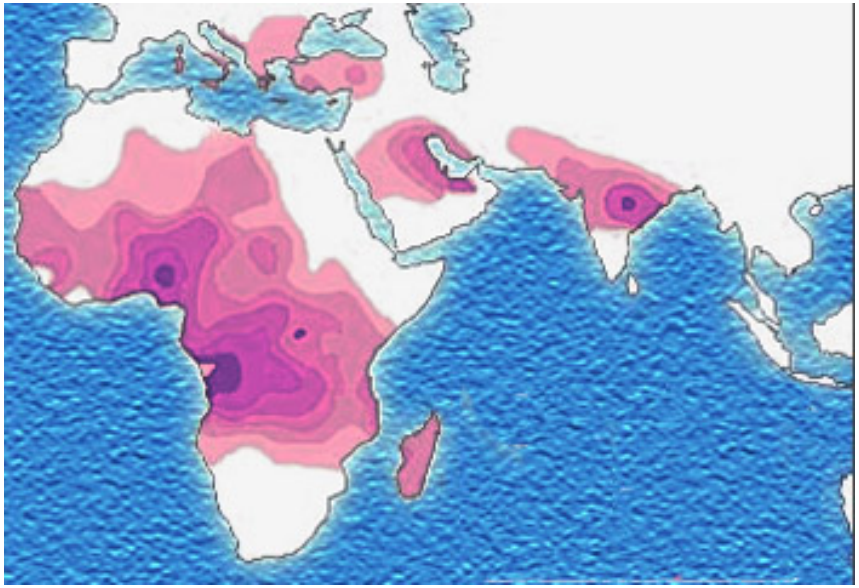
Sickle-Cell Prevalence

- Thus, in malaria regions there is a balance between the fitness gain from helping to fight malaria and the fitness loss due to the offspring that inherit two copies of the mutation



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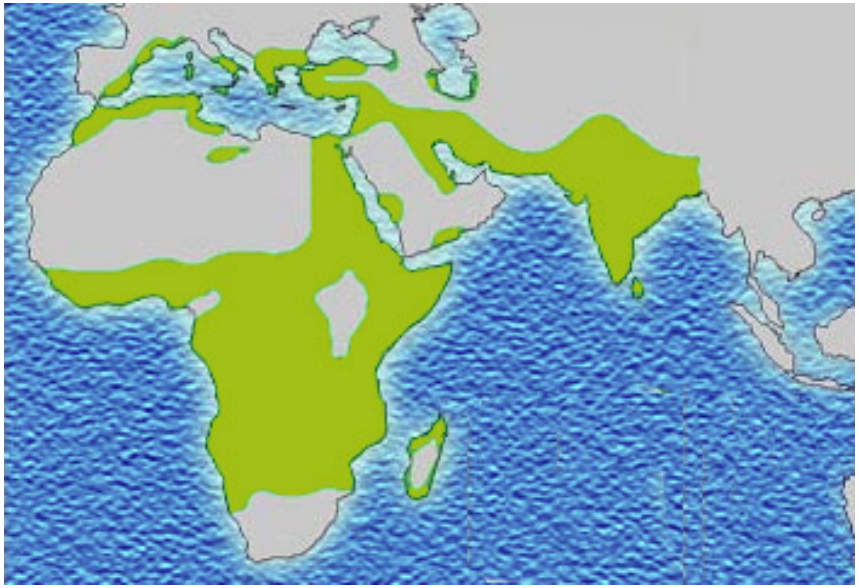

Historic Malaria Prevalence

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Conclusion

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- Finite population effects are crucial to understanding evolution
- With modern computers we can now simulate evolutionary models very easily
- Solving mathematical models forces you to understand the mechanisms that drive evolutionary dynamics
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